

Unnamed_Unit

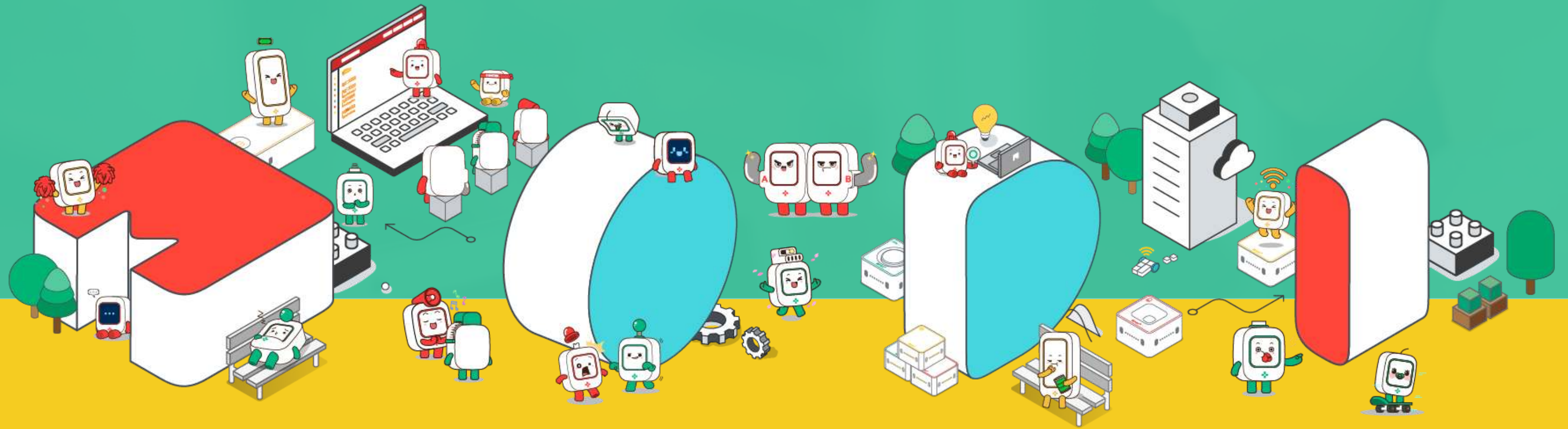
Introduction - Unit 3

PDF Documents

PDF Document 1: 1731918054360-Nanoskool MODI Workshop level2 -1.pdf

This PDF document is included in the unit materials.

YouTube Links



MODI Workshop *with* **MODI+** *with*

NANO
nanoskool.com **SKOOL**



Welcome: Meet the World of STEAM

WELCOME TO THE WORLD OF STEAM



This course is intended to ignite curiosity and nurture an understanding of vital 21st-century skills. We'll begin with the essentials of software and hardware, delve into the digital revolution that sculpted our contemporary world, and then, become inventors in the realm of robotics!



Course Objectives

This MODI workshop is intended to empower students to understand and apply robotics, programming, and STEAM concepts to real-world problems, enhancing their critical thinking and creativity.

By navigating through the course, **students will:**

- ✓ **Acquire hands-on robotics and programming skills with MODI kits.**
 - ✓ **Explore STEAM concepts through cross-disciplinary learning.**
 - ✓ **Utilize programming skills to address real-world problems.**
 - ✓ **Develop critical thinking and creativity.**
-

MODI curriculum aligns with:



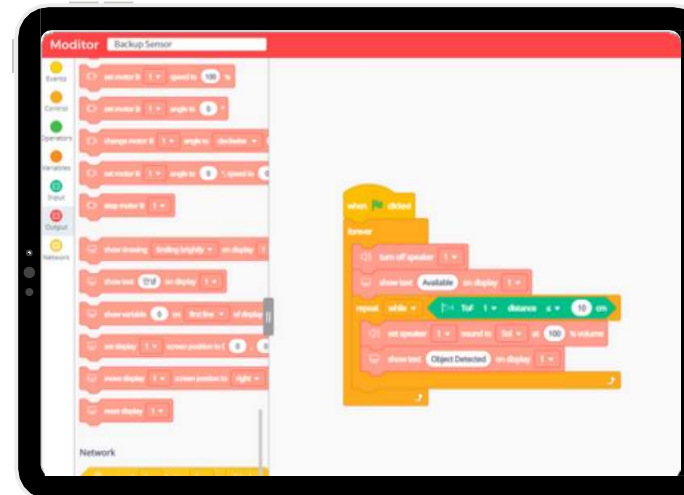
Classroom Tools



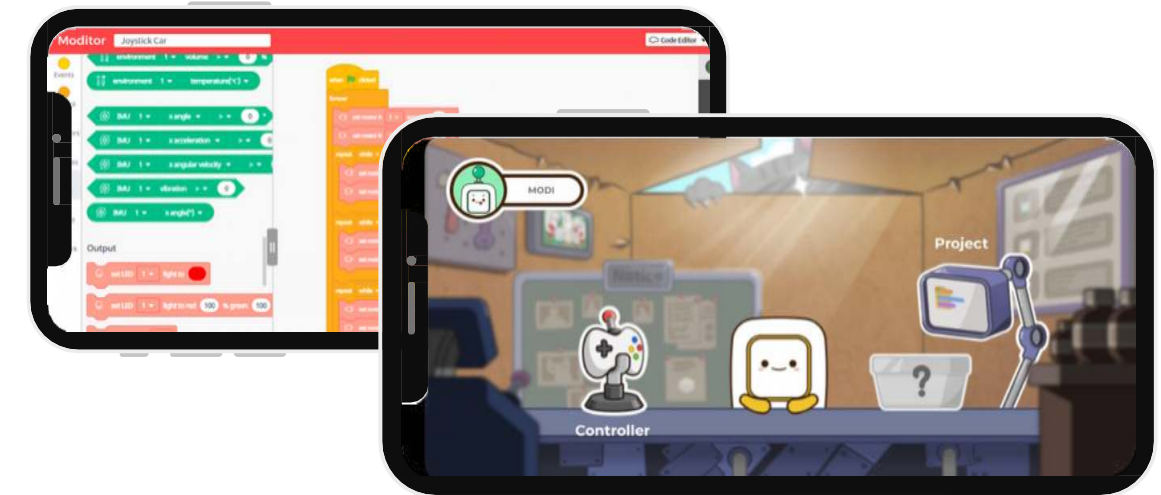
To make the most of our MODI experience,
you'll need one of these electronic devices: a **tablet**, **PC**, or **smartphone**



Pc



t a b l e t



s m a r t p h o n e





Lesson Presentation

Introduction to MODI



What is MODI?

MODI is an exciting tool designed to bring the worlds of coding and robotics to life in a fun and interactive way. It is a set of smart modules that can be connected together to build various robotic configurations.



Critical Thinking Skills

: problems that require logical reasoning and creativity.



Cross-disciplinary Learning

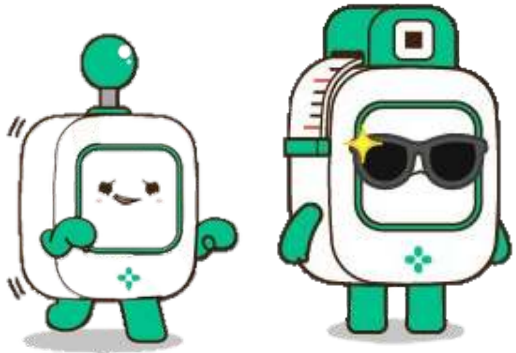
: science, technology, engineering, arts, and mathematics.



Ease of Use

: Simple interface that is intuitive for both teachers and students.





Input modules receive data from the outside and transfer the data to the mainboard.

Input Modules



Button

Senses the status of the button



Environment

Senses temperature, humidity, illuminance, and sound



ToF

Senses distance



Dial

Senses the rotation of the dial



IMU

Senses slopes (Angles)



Joystick

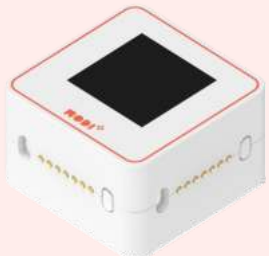
Senses the direction of the stick

Output Modules



Makes sounds

Speaker



Shows letters and images

Display



Motor A, B

Rotates



LED

Em its light



Don't forget!

The output modules receive processed data or instructions from the input modules and translate them into real-world actions.

These actions include:



moving parts !



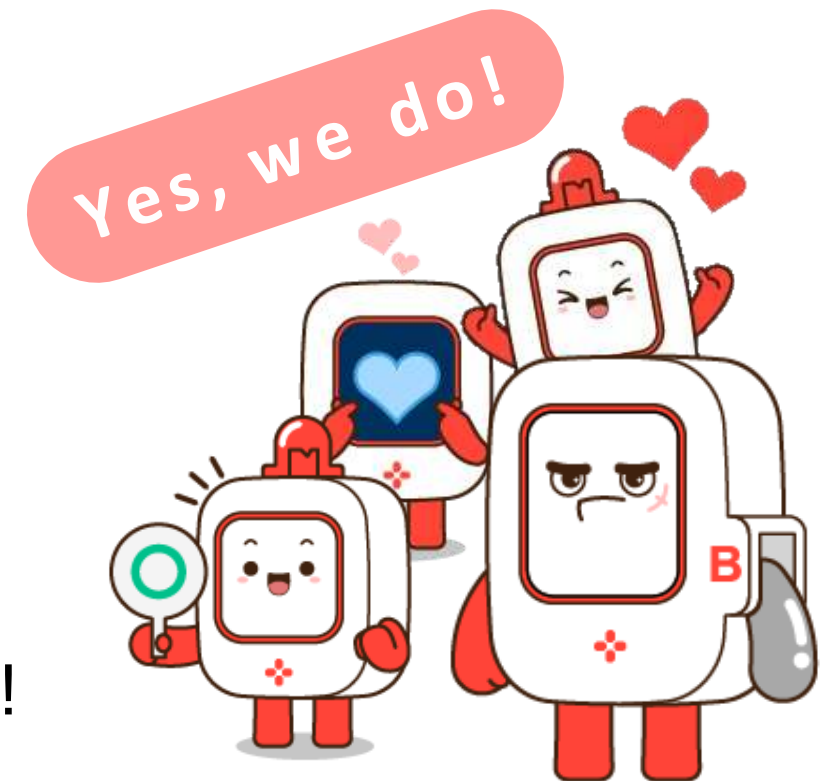
lighting up !



making sounds !



displaying information !





We (setup modules) help all the MODI modules work as intended!!

Setup Modules



Battery

Supplies energy to the modules



Network

Connects modules to other modules, PCs, tablets, and phones

Pop Quiz

What is an example of an output module action?

- a) Receiving data from the outside
- b) Transferring data to other modules
- c) translating data into real-world actions
- d) Supplying power to the modules

Discussion

Explain in your own words the relationship between input and output modules in MODI?



Introduction to MODI

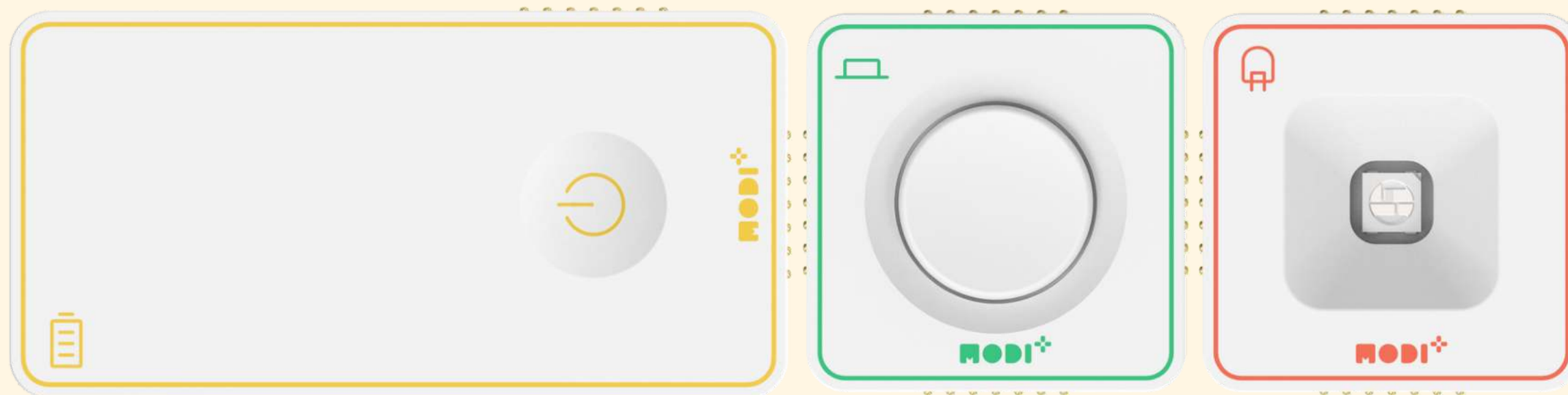
Introduction to MODI - Part.1

UNPLUGGED ACTIVITIES

Let's get ready for some fun and engaging MODI unplugged activities!
These activities will help us understand coding and robotics concepts without needing a computer or tablet.

Unplugged Activity #1 - Flashlight!

Connect the Button module and the LED module in any order or orientation!



Introduction to MODI

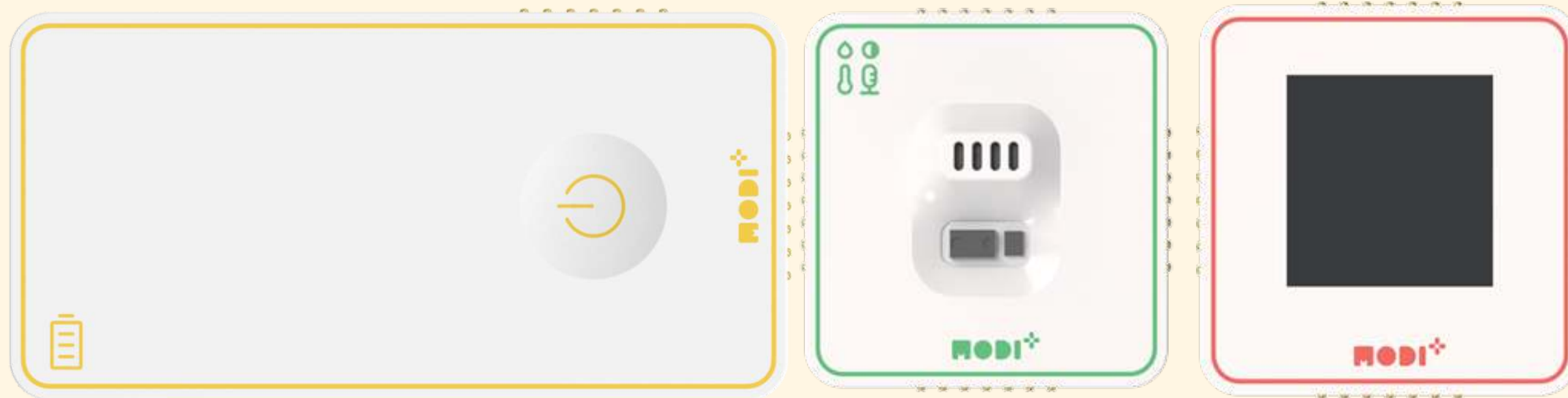
Introduction to MODI - Part.1

UNPLUGGED ACTIVITIES

The environment module reads the surrounding temperature, humidity, sound level, and illuminance. The display module displays the information read by the environment module.

Unplugged Activity #2 – Environment monitor

Read the information around you out loud.



Let's make an LED speaker with a dial module!

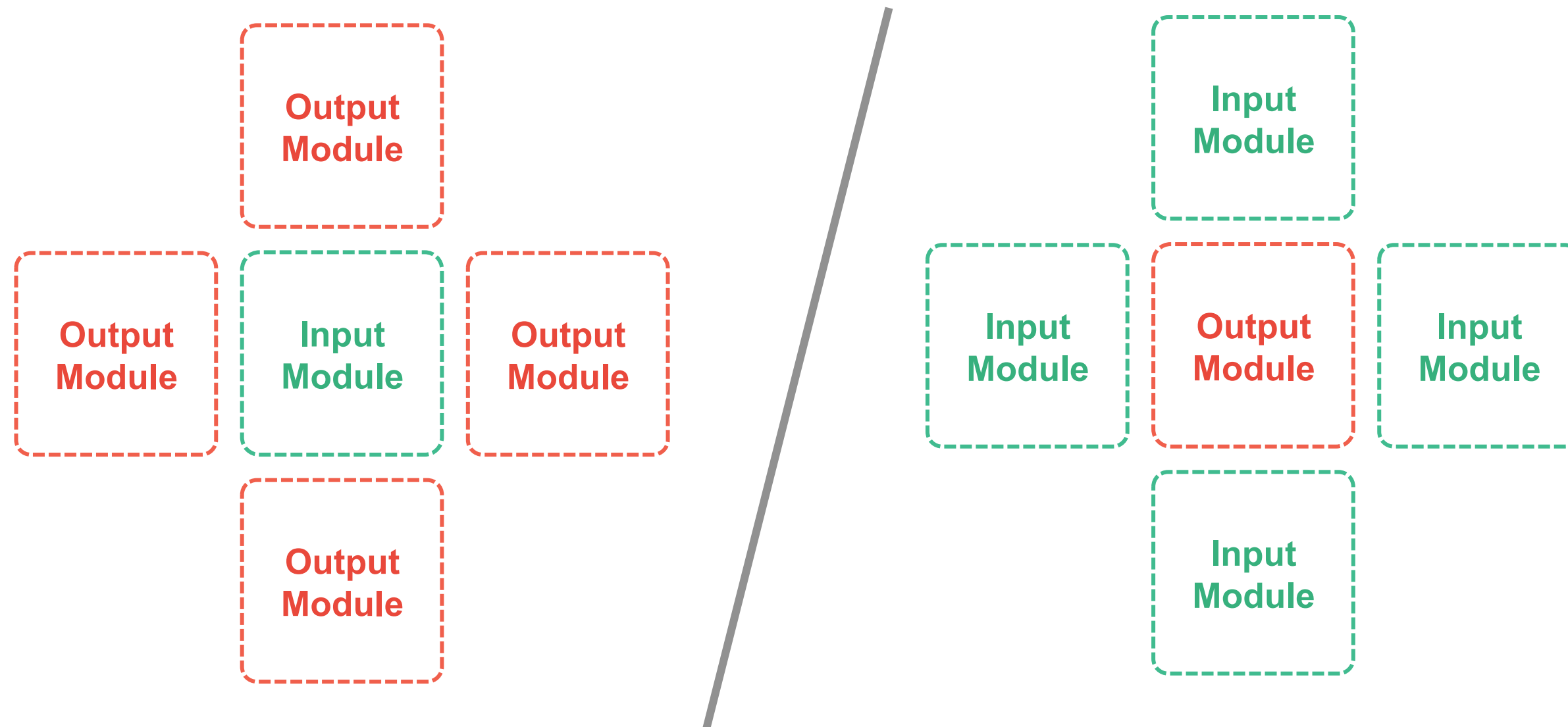
Unplugged Activity #3

The input module can control more than 2 modules simultaneously.

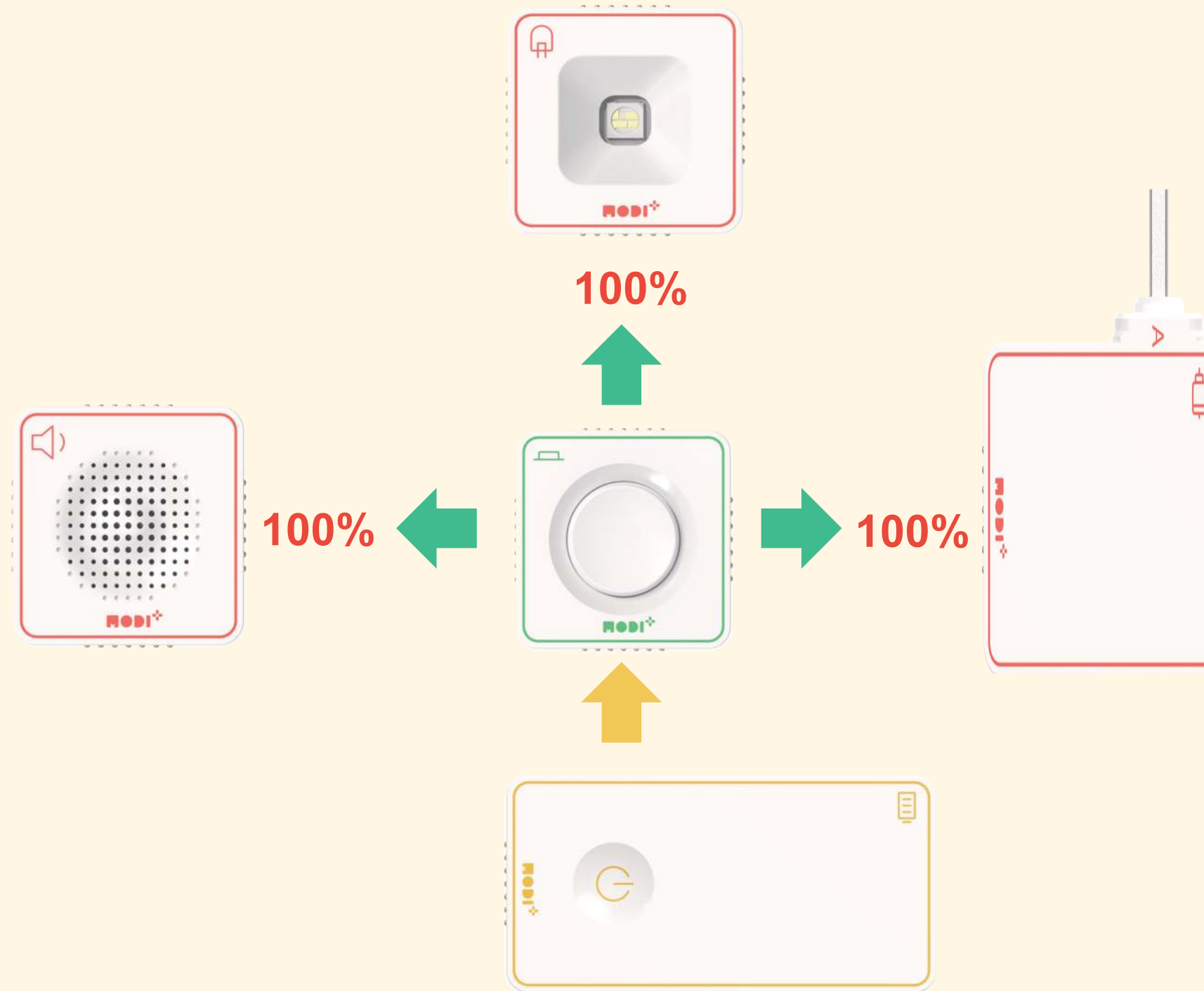


Let's figure out what happens with the following.

1. When multiple output modules are connected to an input module.
2. When multiple input modules are connected to an output module.



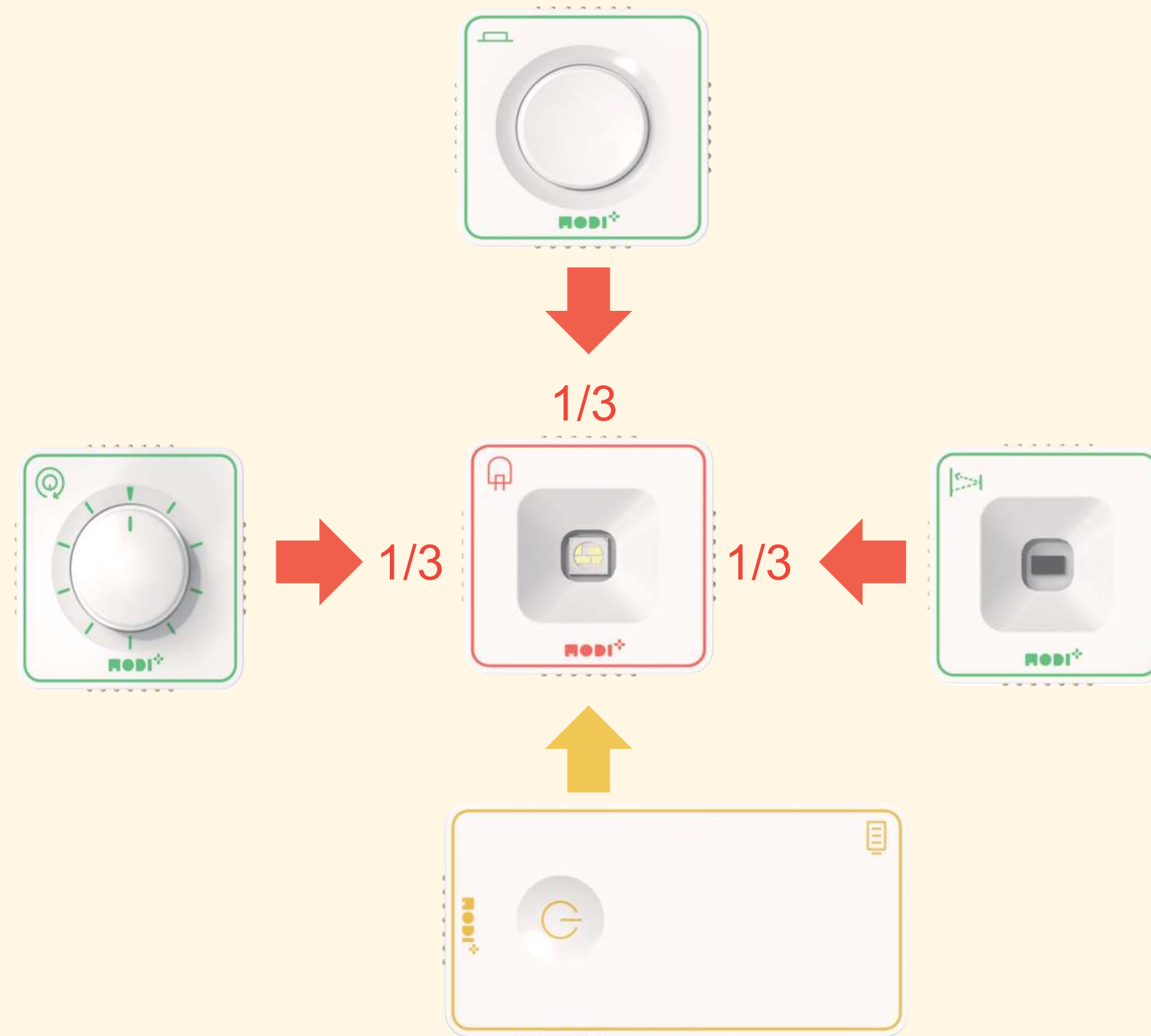
UNPLUGGED ACTIVITIES



An input module affects all adjacent output modules.



UNPLUGGED ACTIVITIES



An output module is affected by adjacent output modules equally.





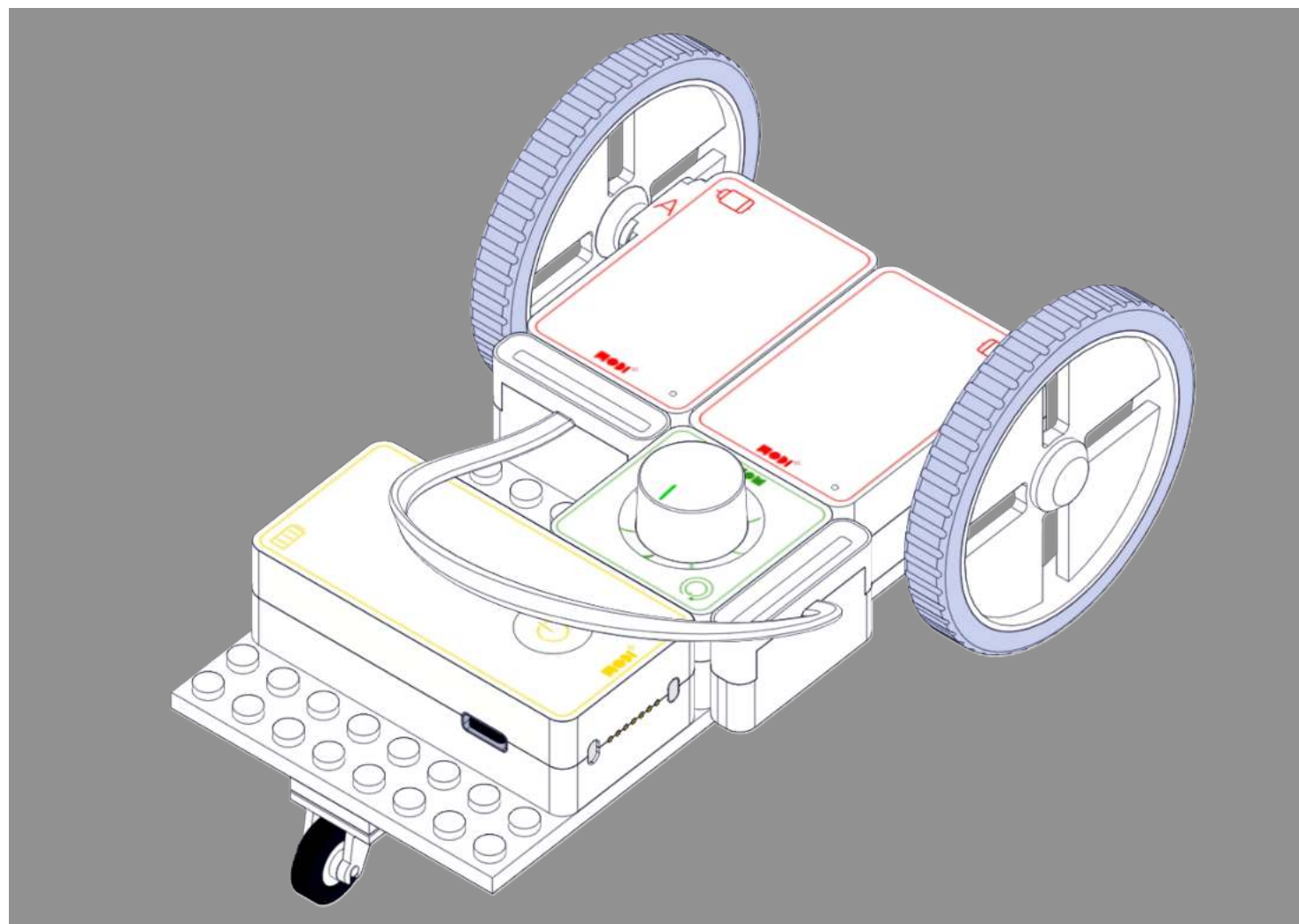
How many modules are
in this image?



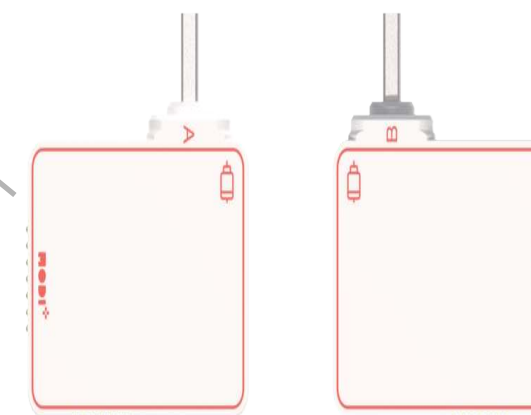
Each part of the module
can be connected with the
magnetic side!



UNPLUGGED ACTIVITIES



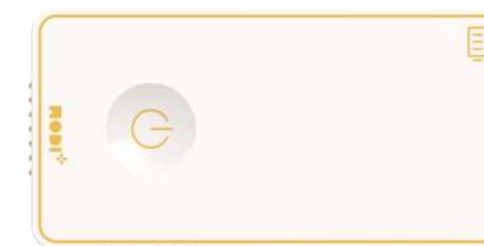
**To make the car
move**



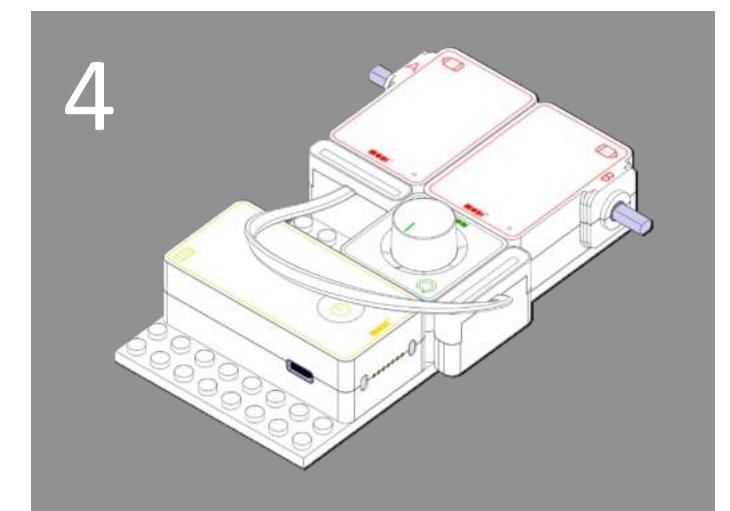
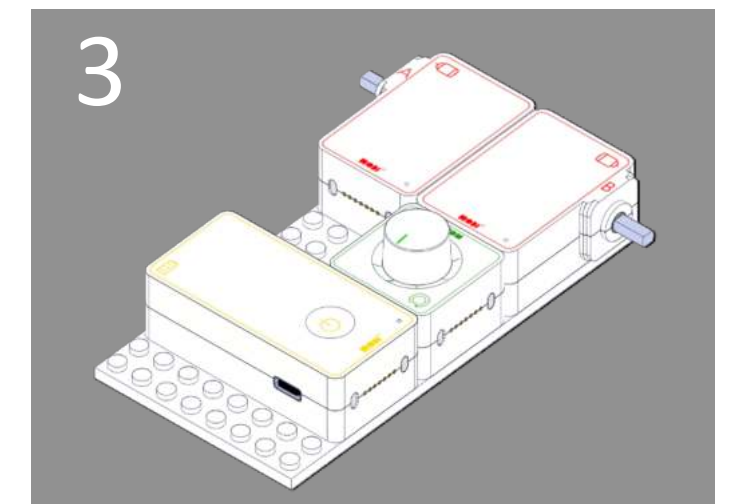
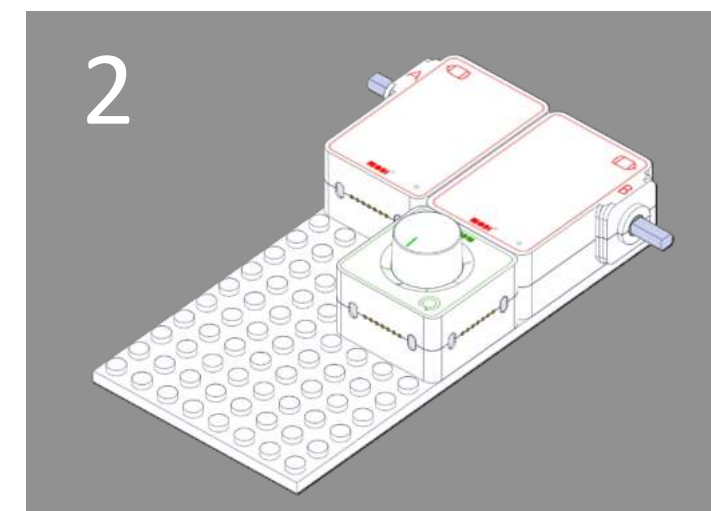
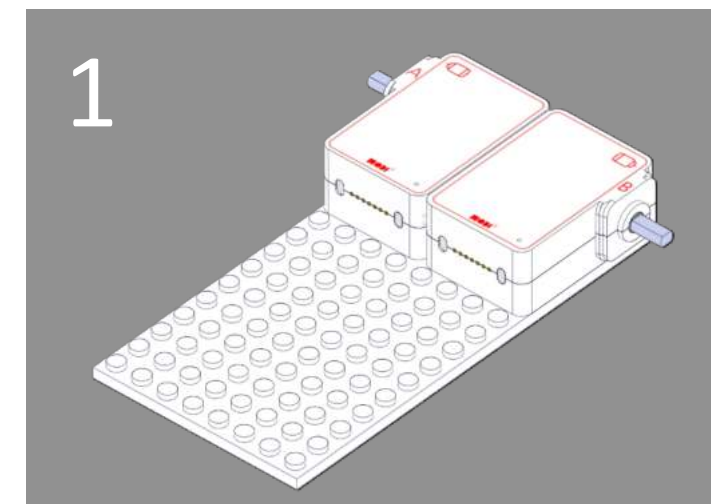
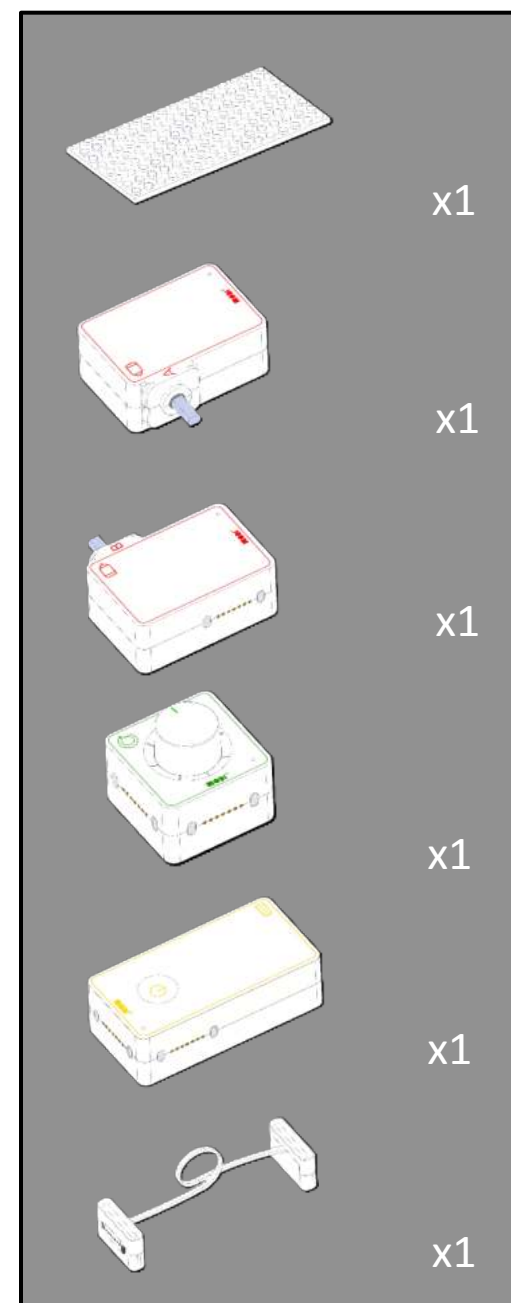
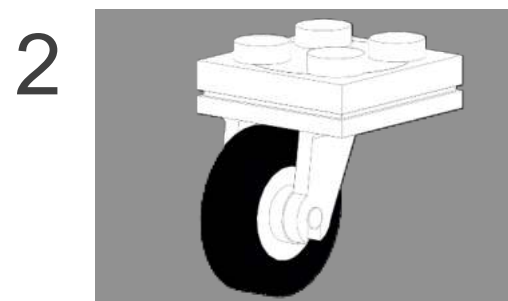
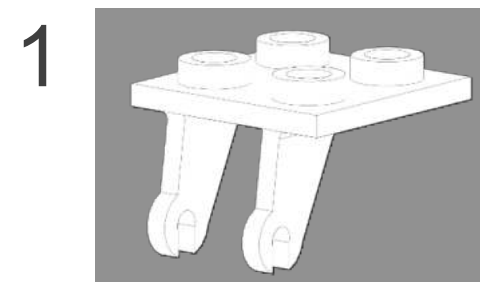
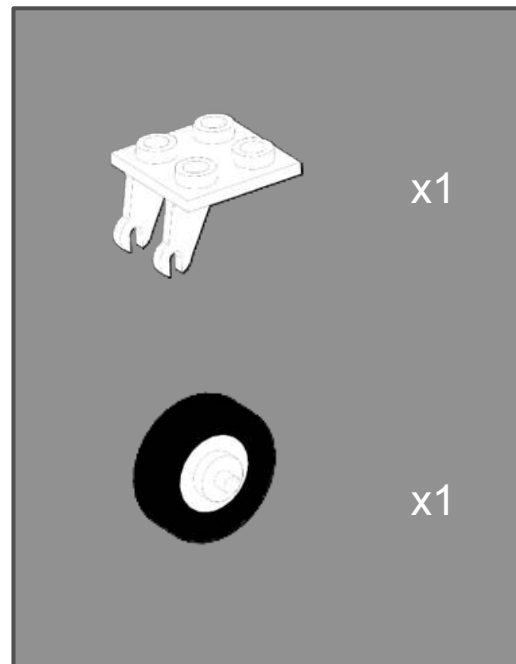
**To control the speed of th
e car**



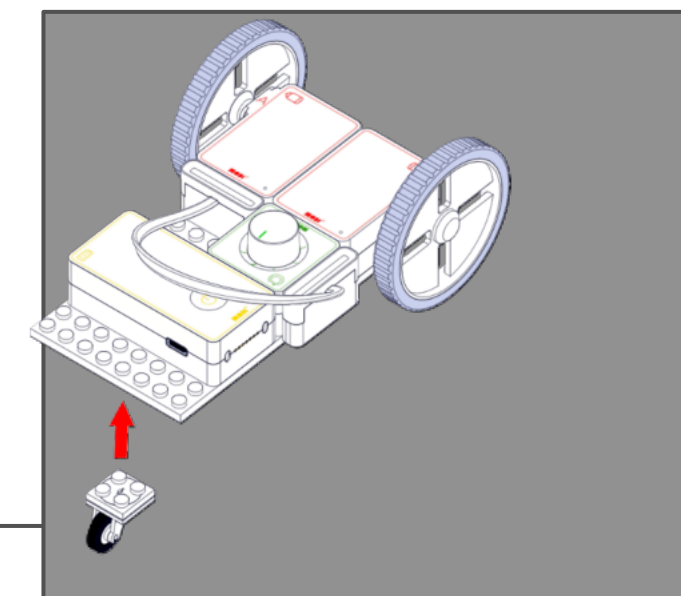
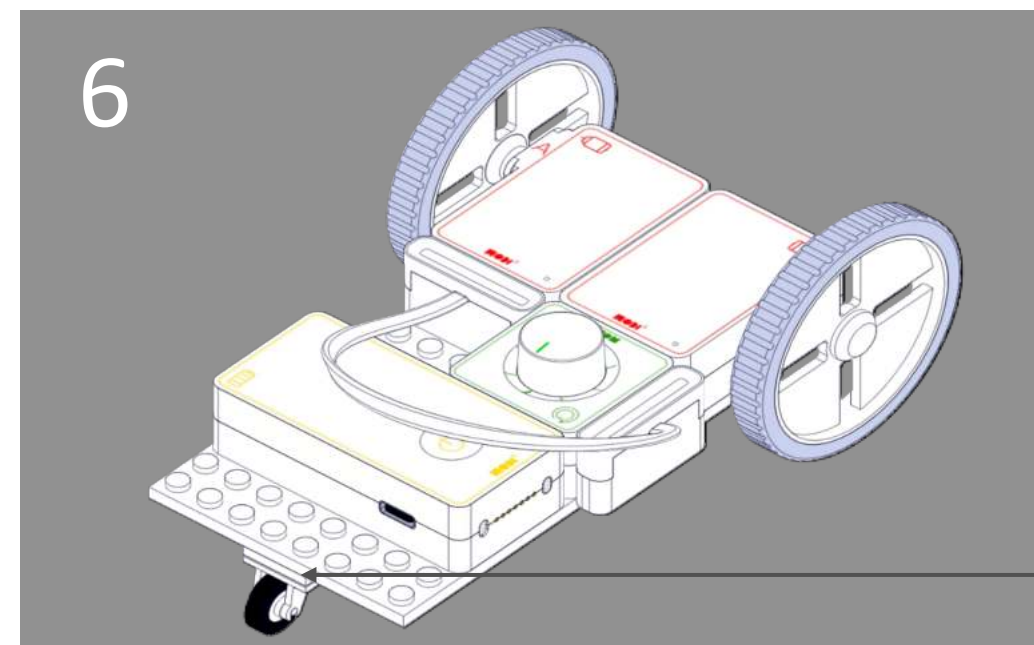
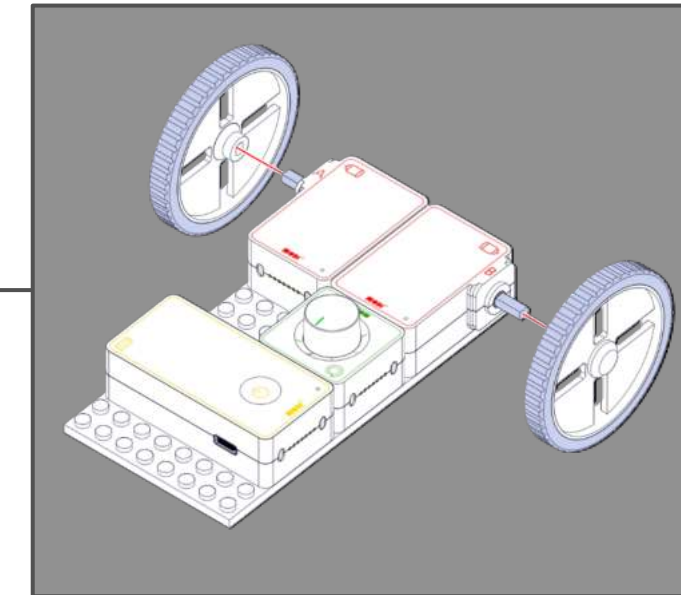
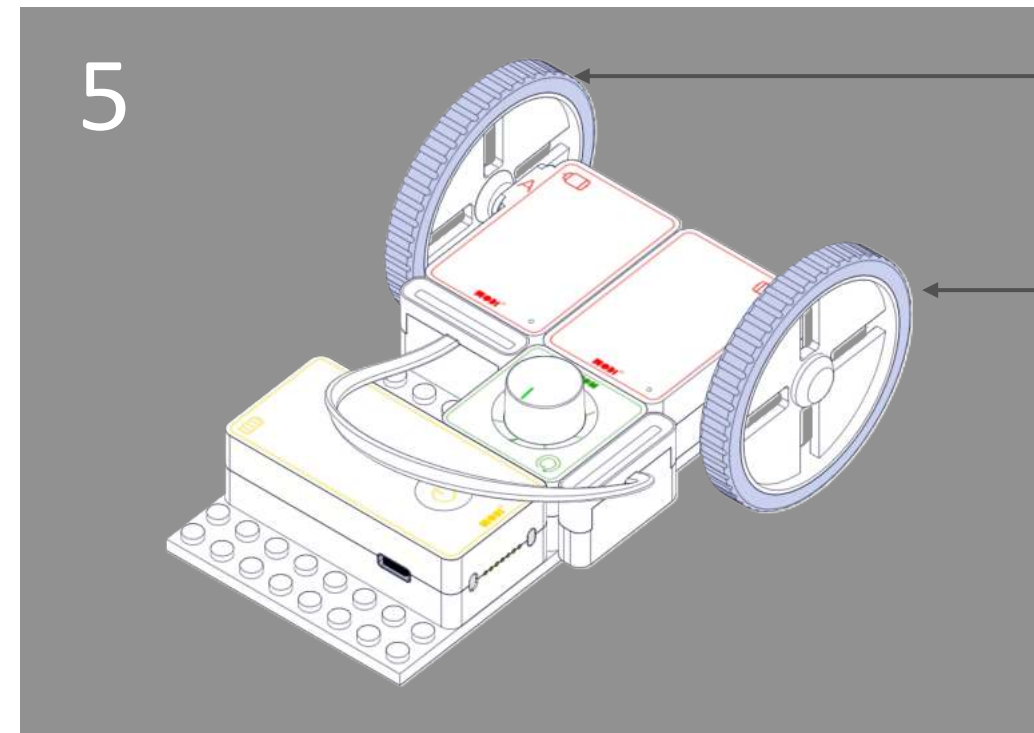
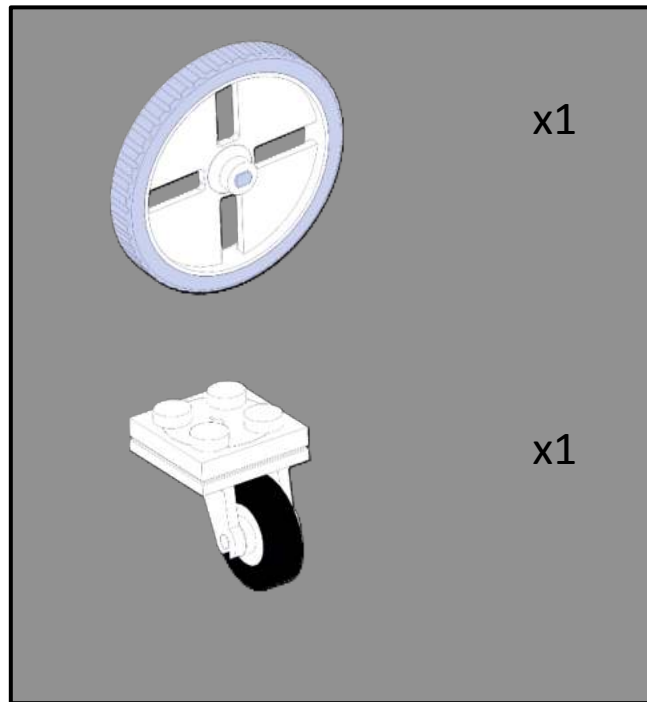
**To supply the
energy**



UNPLUGGED ACTIVITIES

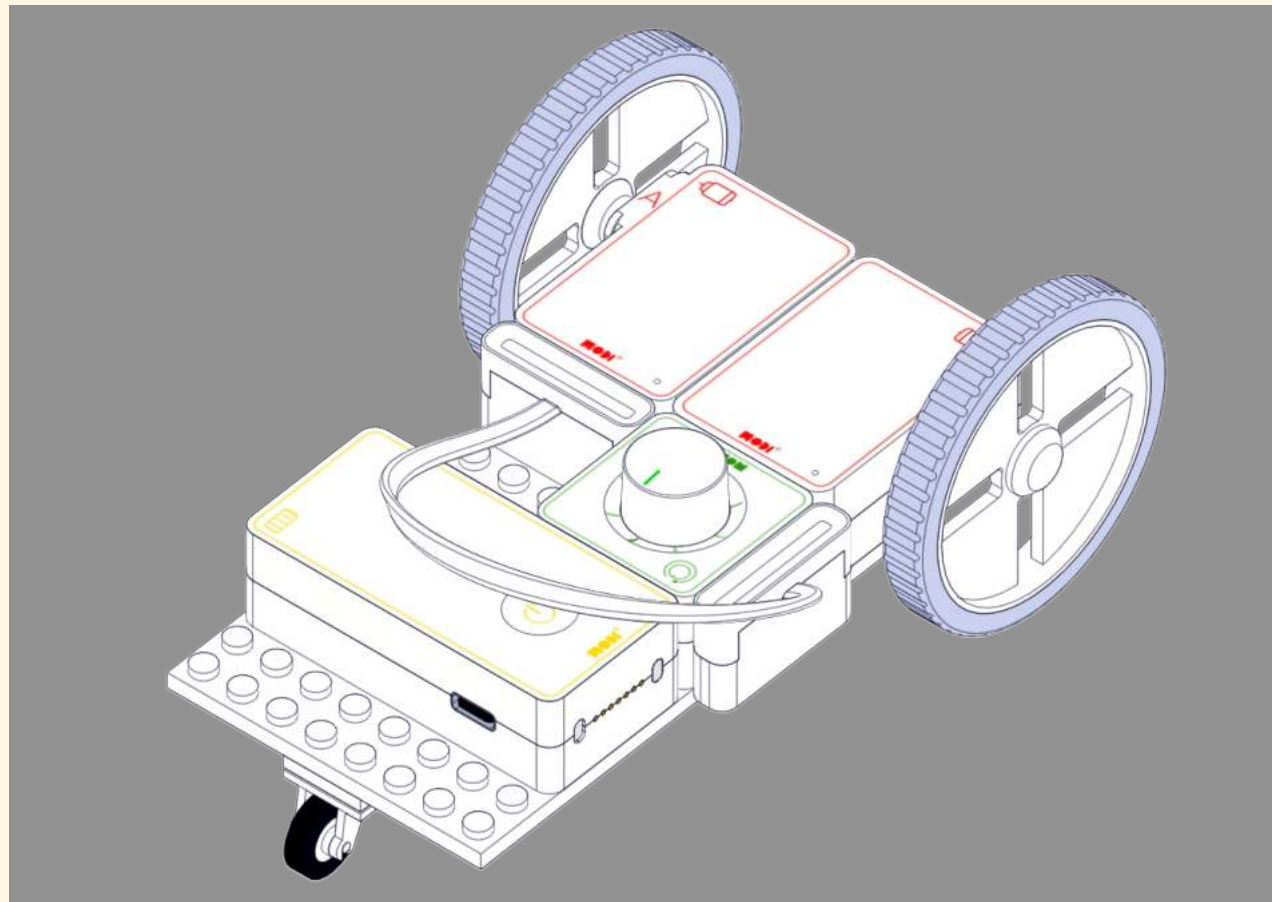


UNPLUGGED ACTIVITIES



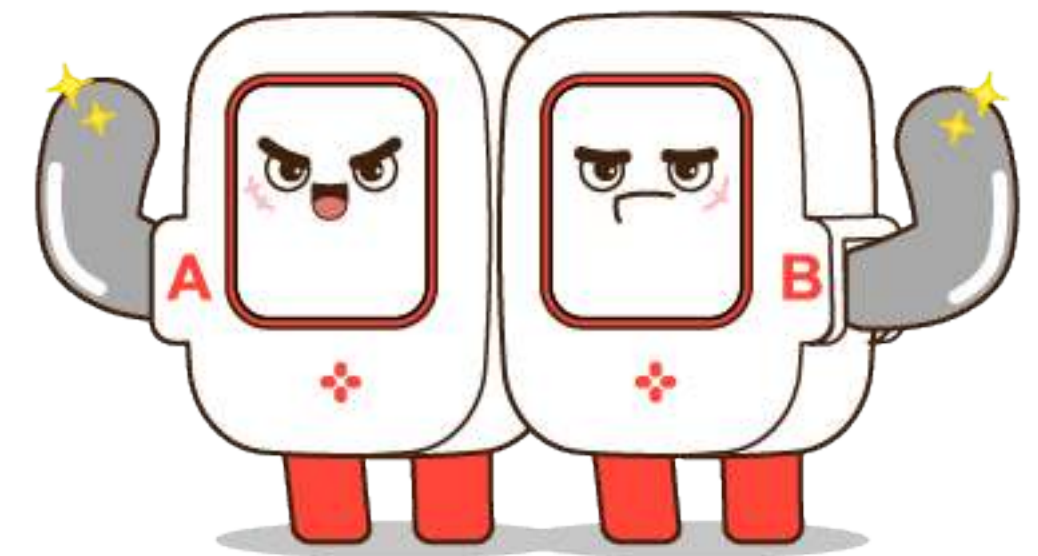
Unplugged Activity #4 – Car

We used the Dial module to control the speed of the car.



Discussion

- Why did we use a module connector?
- Without module connector what will happen?



- ✓ **Mission**
- Change the structure of the car to add an LED module.
 - With one dial, control the speed of the car and the brightness of the LED

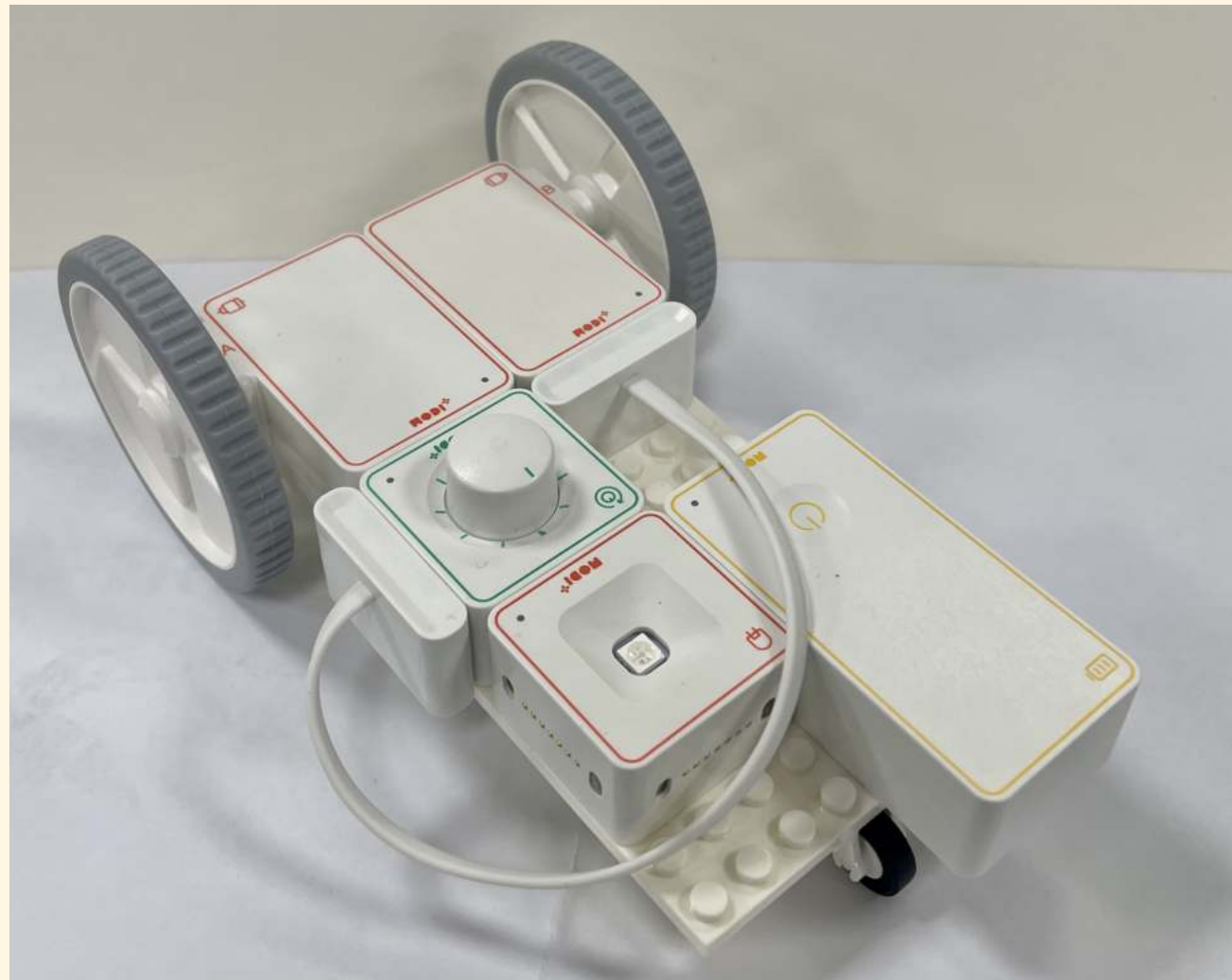
Unplugged Activity #3 – LED Car

ANSWER IS..

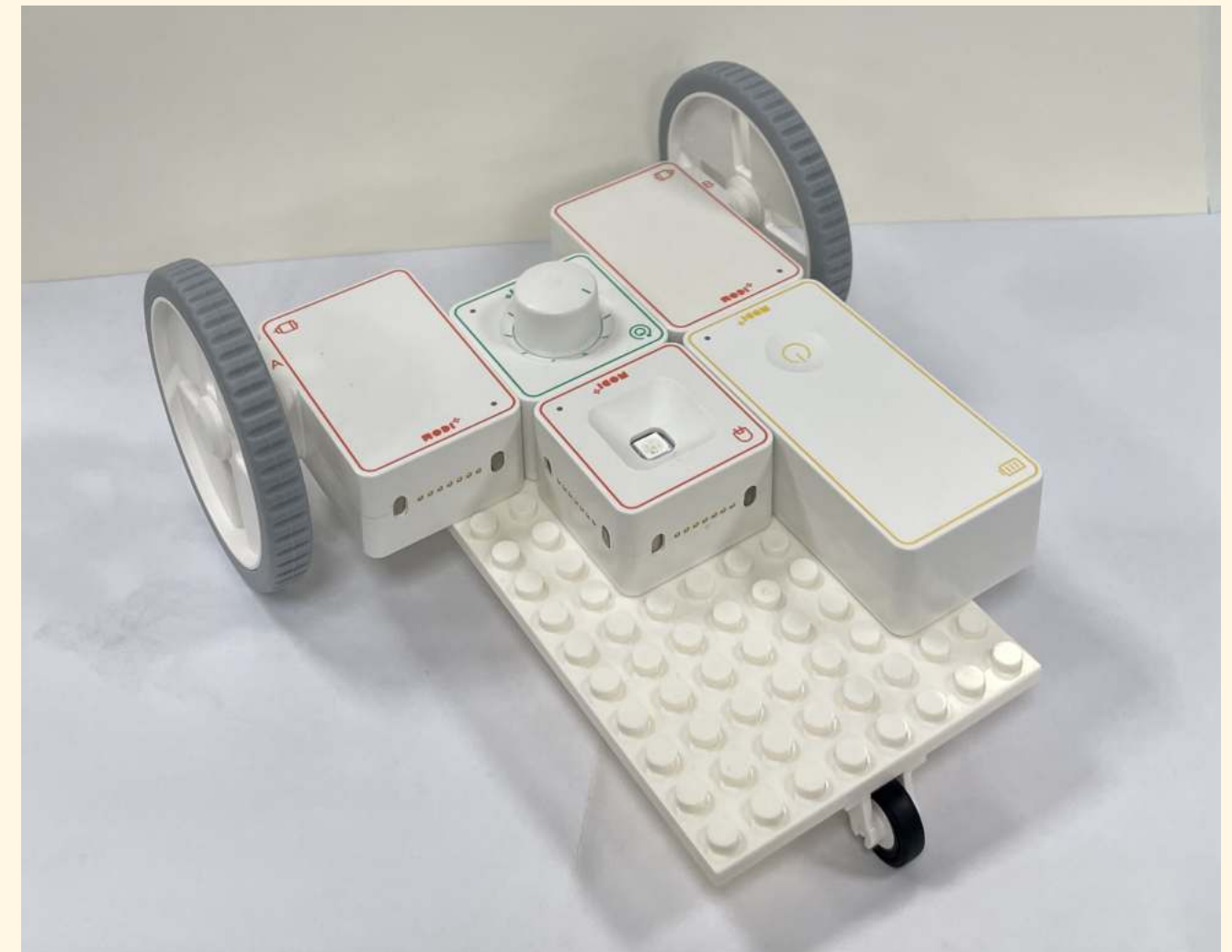
- ✓ **Mission**
- Change the structure of the car to add an LED module.
 - With one dial control the speed of the car, and the brightness of the LED

Unplugged Activity #3 – LED Car

✓
Type1



✓
Type2



Takeaway Points

- ✓ We learned about the terms "input" and "output" in the context of computing.
- ✓ We understood how these concepts of input and output apply to MODI modules.
- ✓ We examined the potential creations we can make using MODI modules.

You're on your way to becoming inventors of
the 21st century!





Lesson Presentation

Introduction to Code Editor

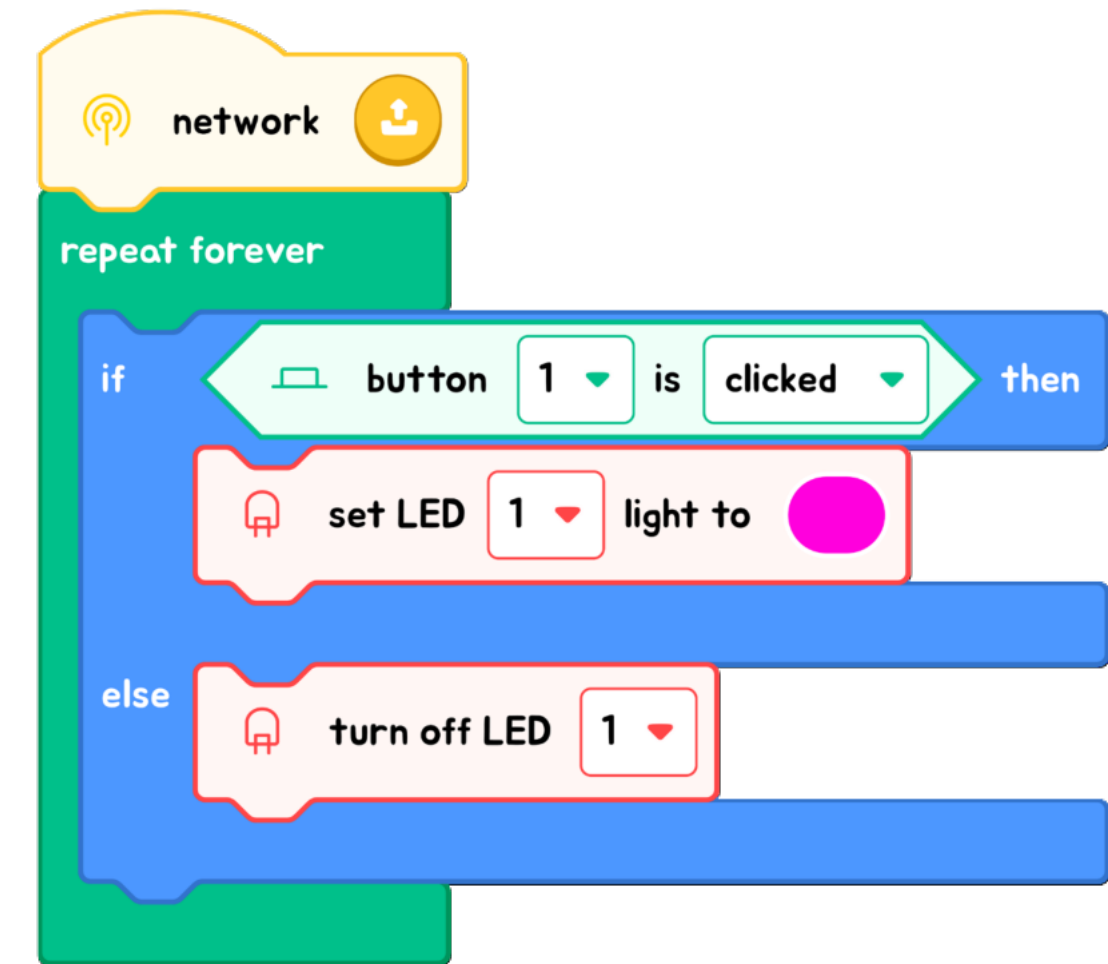
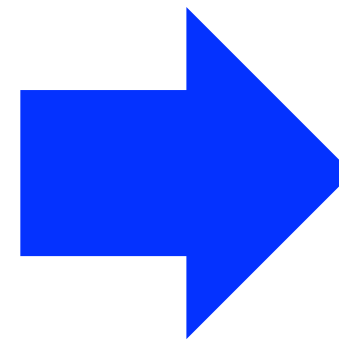


Introduction to Code Editor

Introduction to MODI – Part.2

PROGRAMMING ACTIVITIES

```
1 requests.get(url)
2 # checking response.status_code (if you get 502, try rerunning the code)
3 if response.status_code != 200:
4     print(f"Status: {response.status_code} - Try rerunning the code!")
5 else:
6     print(f"Status: {response.status_code}\n")
7 # using BeautifulSoup to parse the response object
8 soup = BeautifulSoup(response.content, "html.parser")
9 # finding Post images in the soup
10 images = soup.find_all("img", attrs={"alt": "Post image"})
11
```



What is "Coding"?

Coding is like giving instructions to a computer or a robot, telling it what to do step by step. We use special languages called programming languages to write these instructions.

Why Block Coding?

In block coding, each 'block' is a piece of code or a command that tells the computer what to do. Just like toy blocks, we can snap these code blocks together to build a program. For MODI, we are going to use this method of coding!



Easy Start:

Block coding is like a puzzle, without any need to worry about spelling or typing errors.



Visual Aid:

With block coding, you can visually see the flow of your program.



Logic-Focused:

It helps understand core coding concepts like sequences, loops, and conditions.

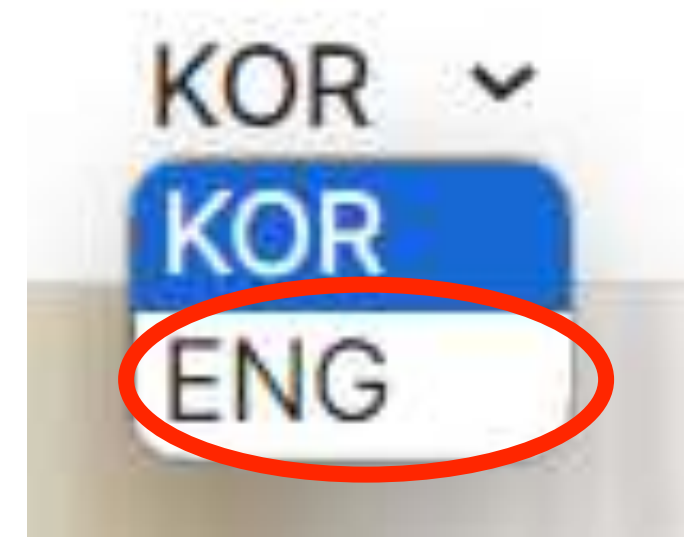
Follow the steps to explore

Let's MODI!

Step 1 Visit 'MODIPlanet' on your browser



Step 2 Click the KOR and change the language as English!

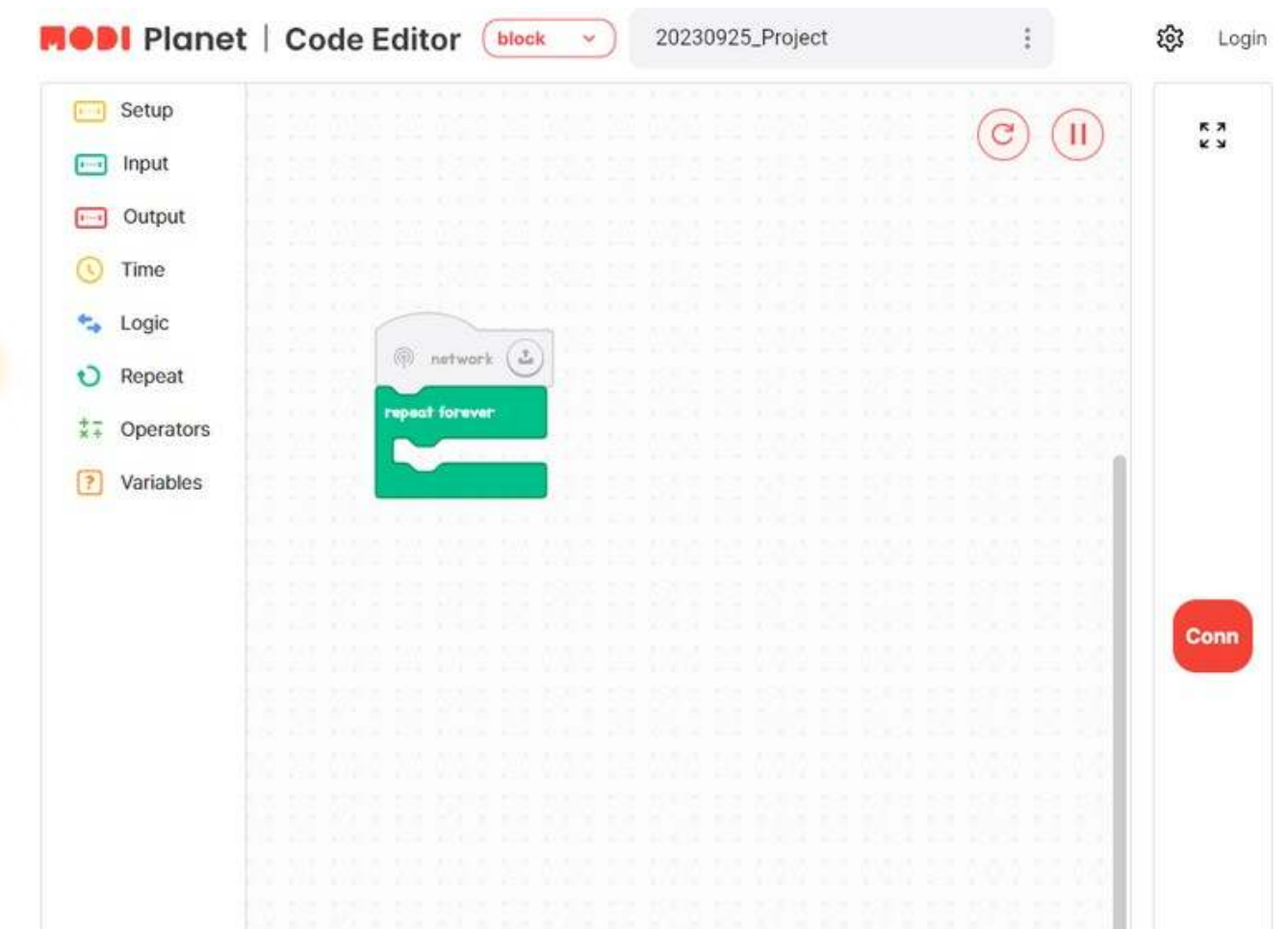
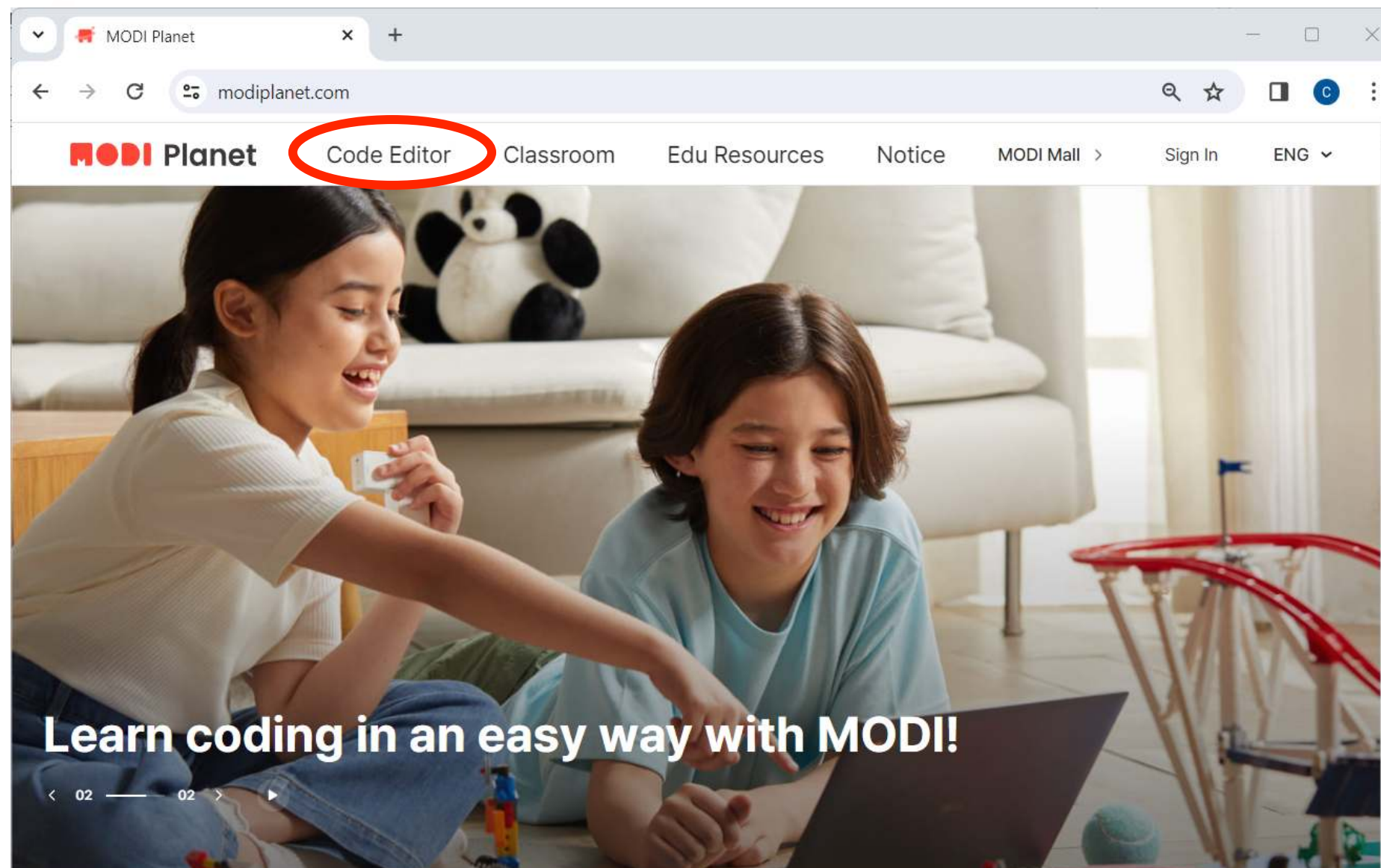


Do not use google translator!



Follow the steps to explore **Let's MODI!**

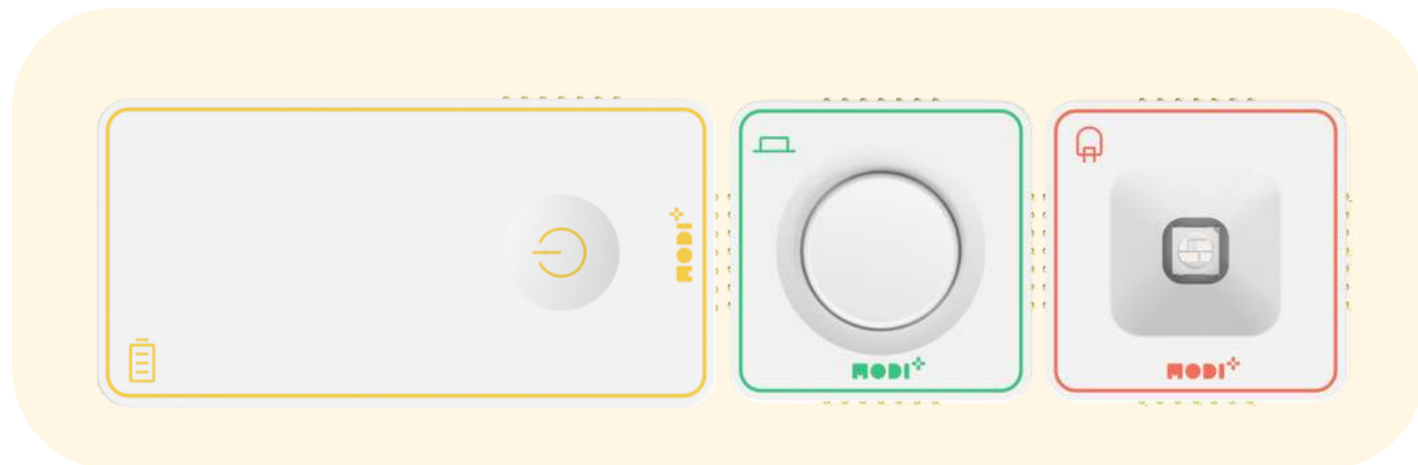
Step 3 Click on 'Code Editor'! Then, you are ready to code!



Basic Programming Activity #1 - "Let's change the color!"

We are going to work on programming the flashlight that we made in our last class.

This will help us understand how to use 'code editor'.

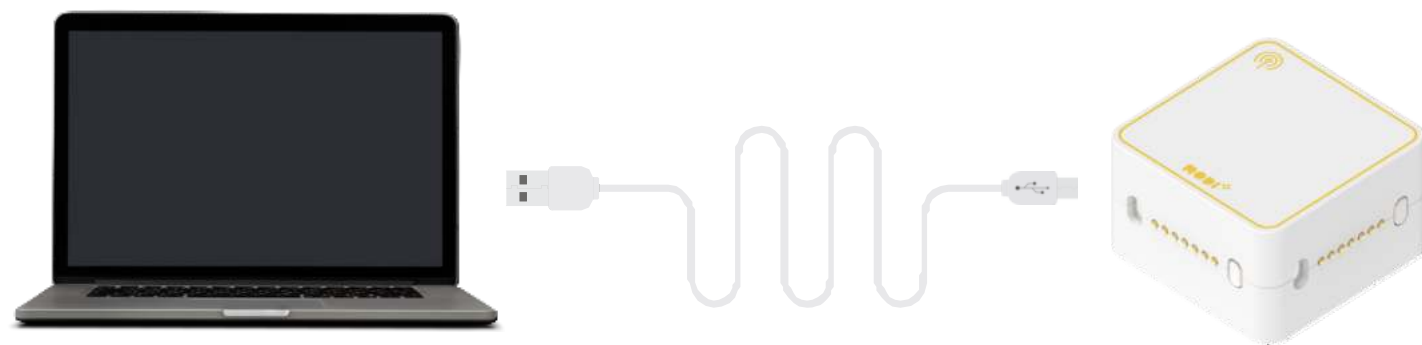


We're going to change the color of the LED light!



Step 1

Connect the network module to your device with the type-c cable in the package



Step 2

Add the button, LED modules to the network.



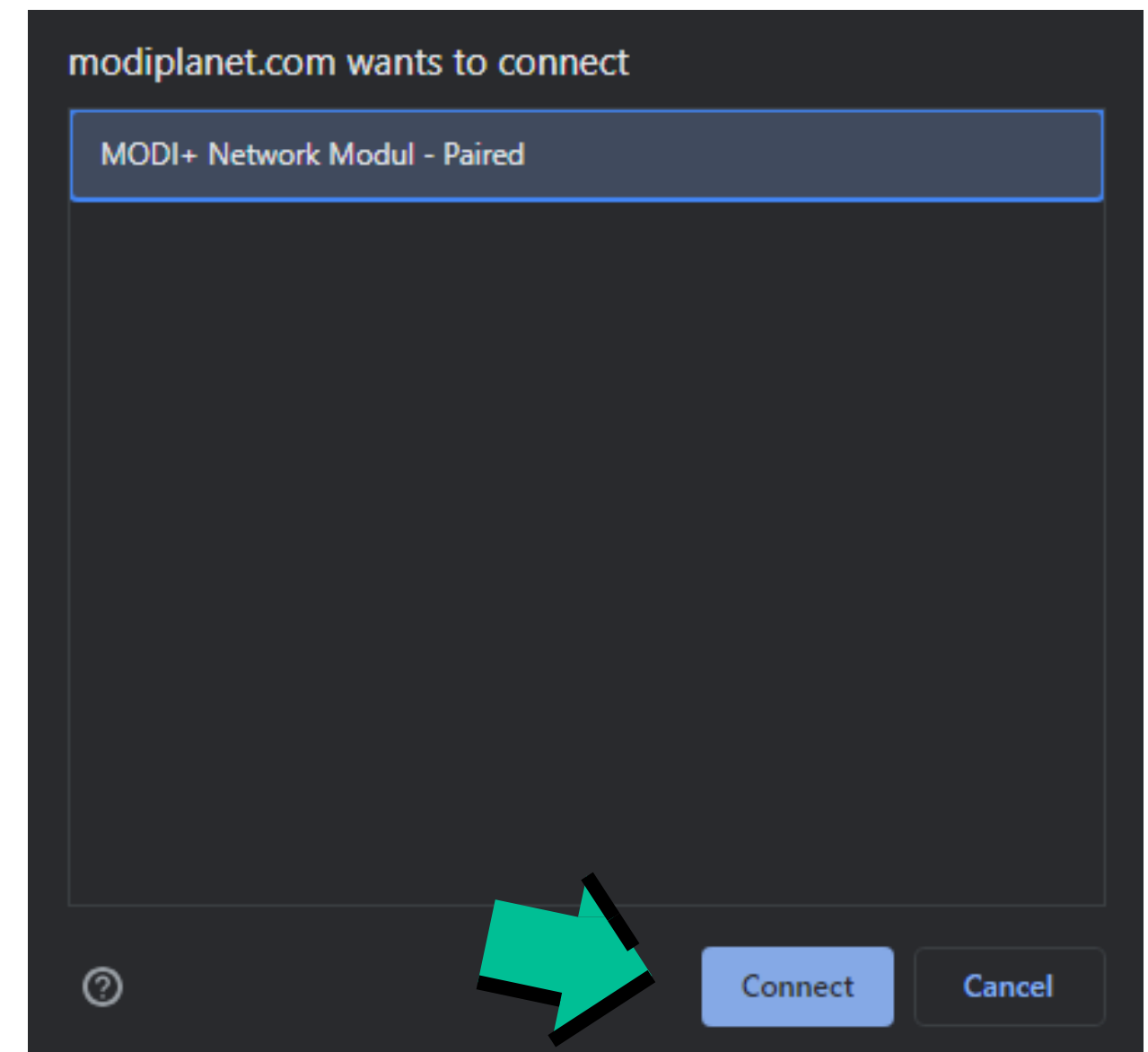
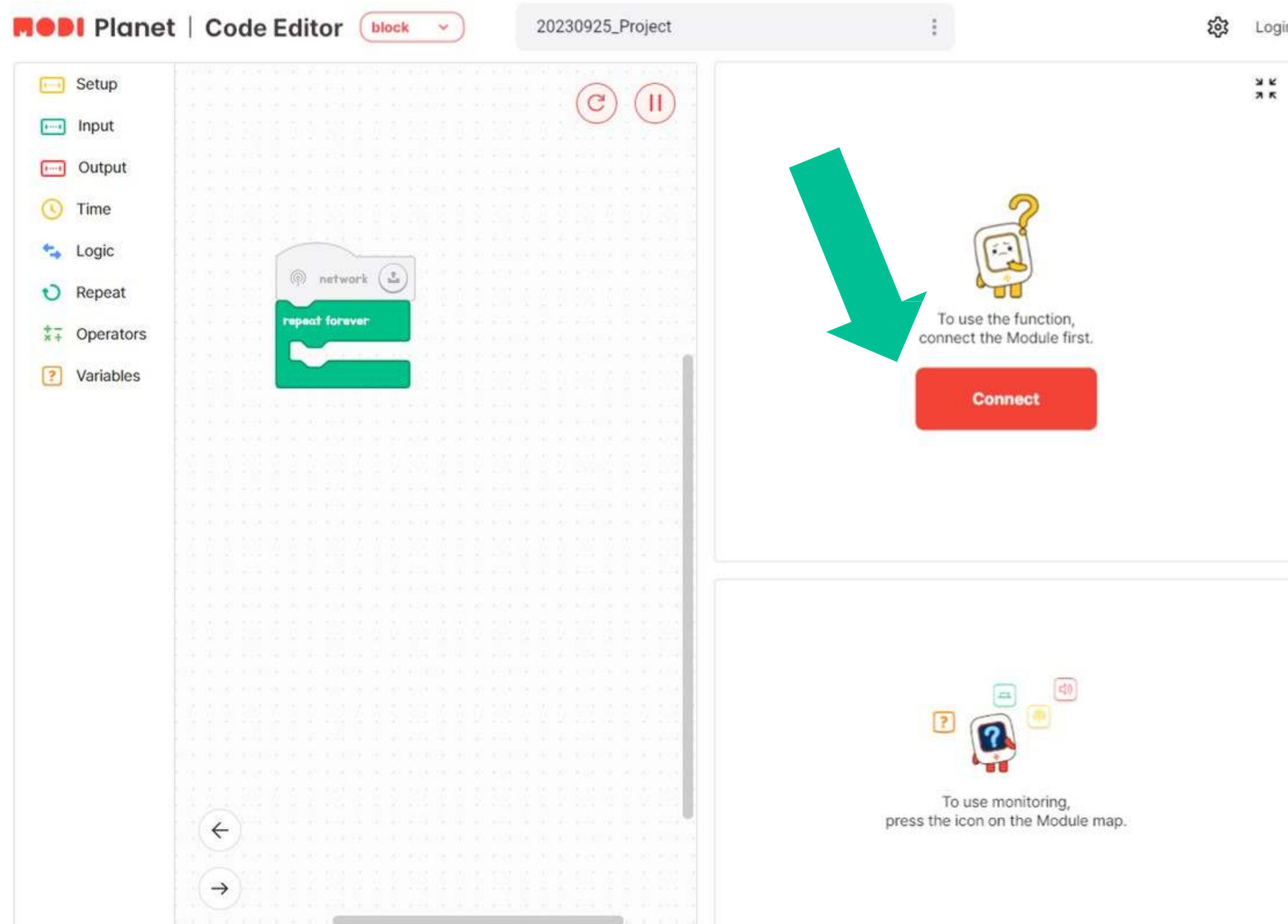
Introduction to Code Editor

Introduction to MODI – Part.2

PROGRAMMING ACTIVITIES

Step 3 Click 'Connect' in Code Editor to pair the network module with the device.

Step 4 Click the network module in the list, and click the 'Connect' button



Step 5

Think about the sequence how the flashlight works!
Each of the blocks in section!

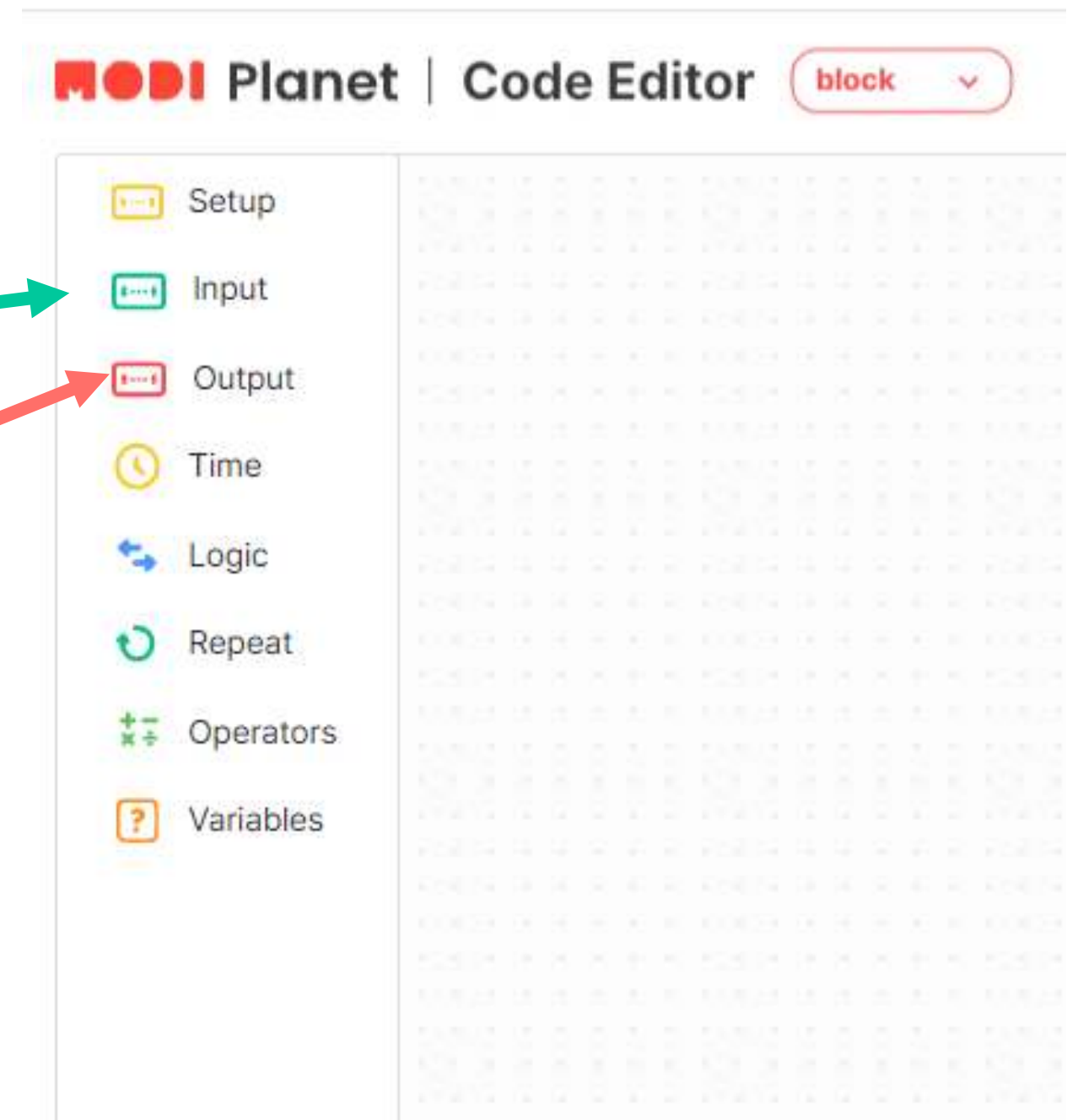
<Algorithm>

When we click the button, the LED is on!

button is clicked



LED is on



Step 6

Now, it is time to drag and drop blocks! By dragging and dropping, complete the code shown below.

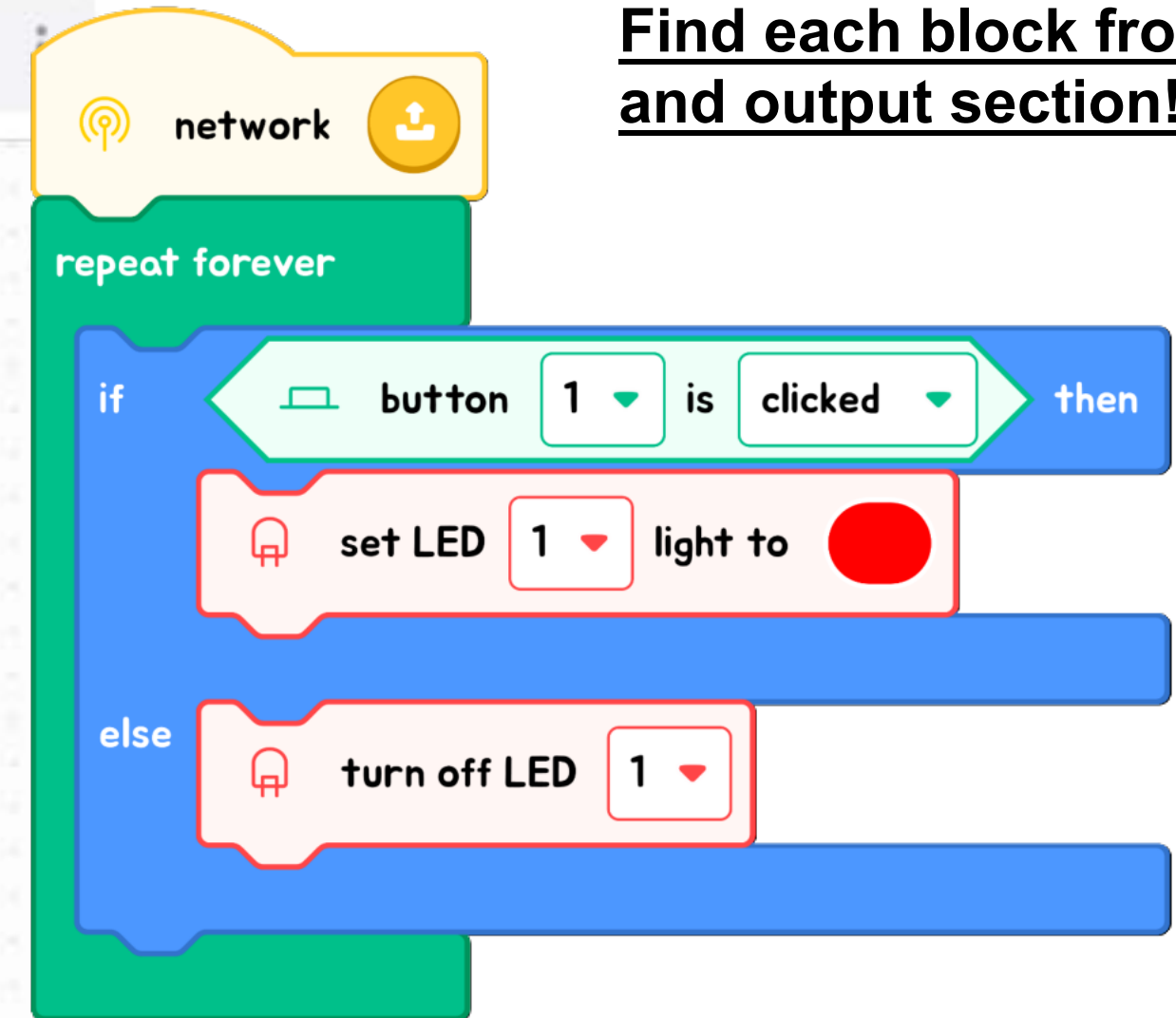
Click the 'Logic' section!

MODI Planet | Code Editor

block

20240114_Project

- Setup
- Input
- Output
- Time
- Logic**
- Repeat
- Operators
- Variables



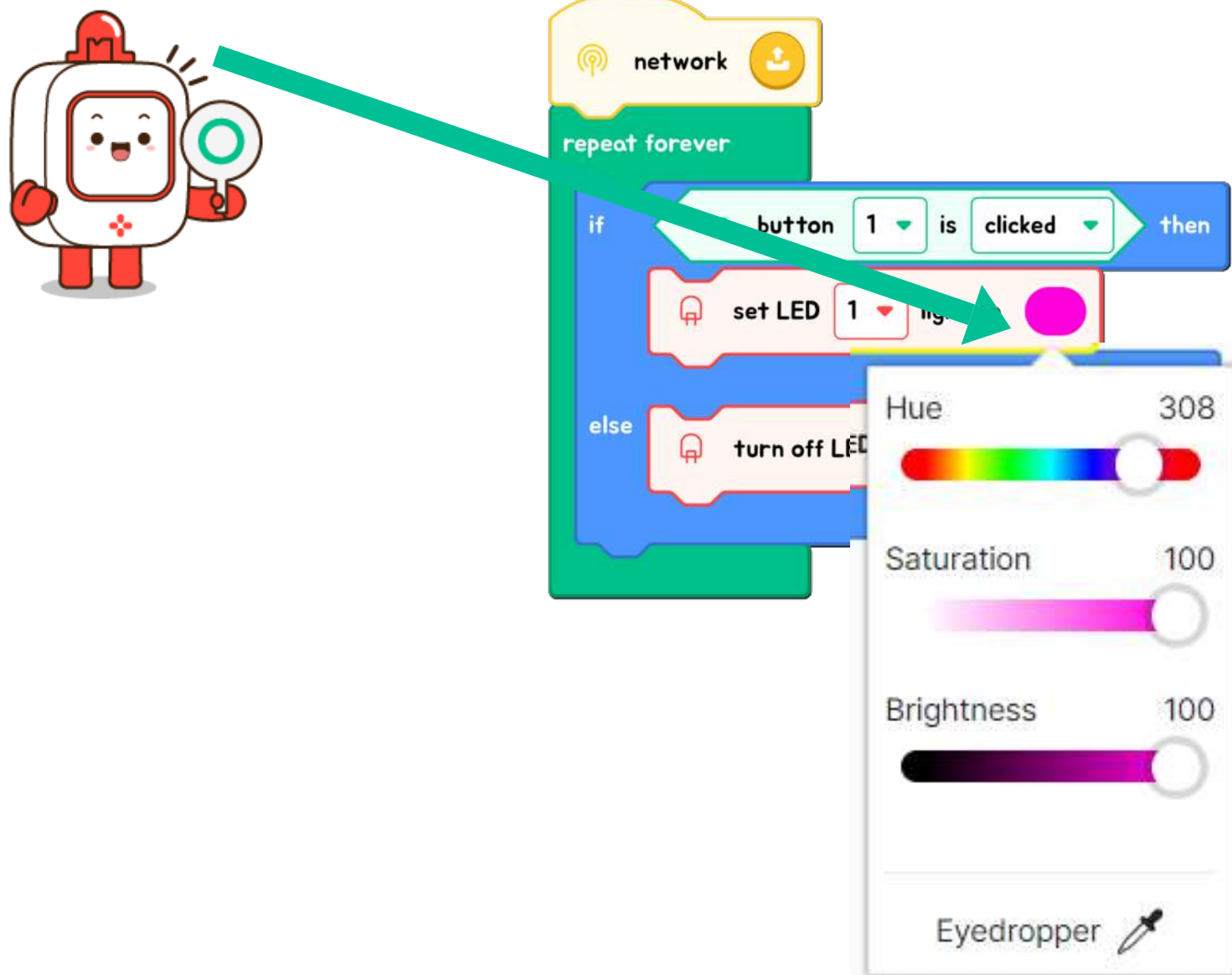
Find each block from input and output section!

Introduction to Code Editor

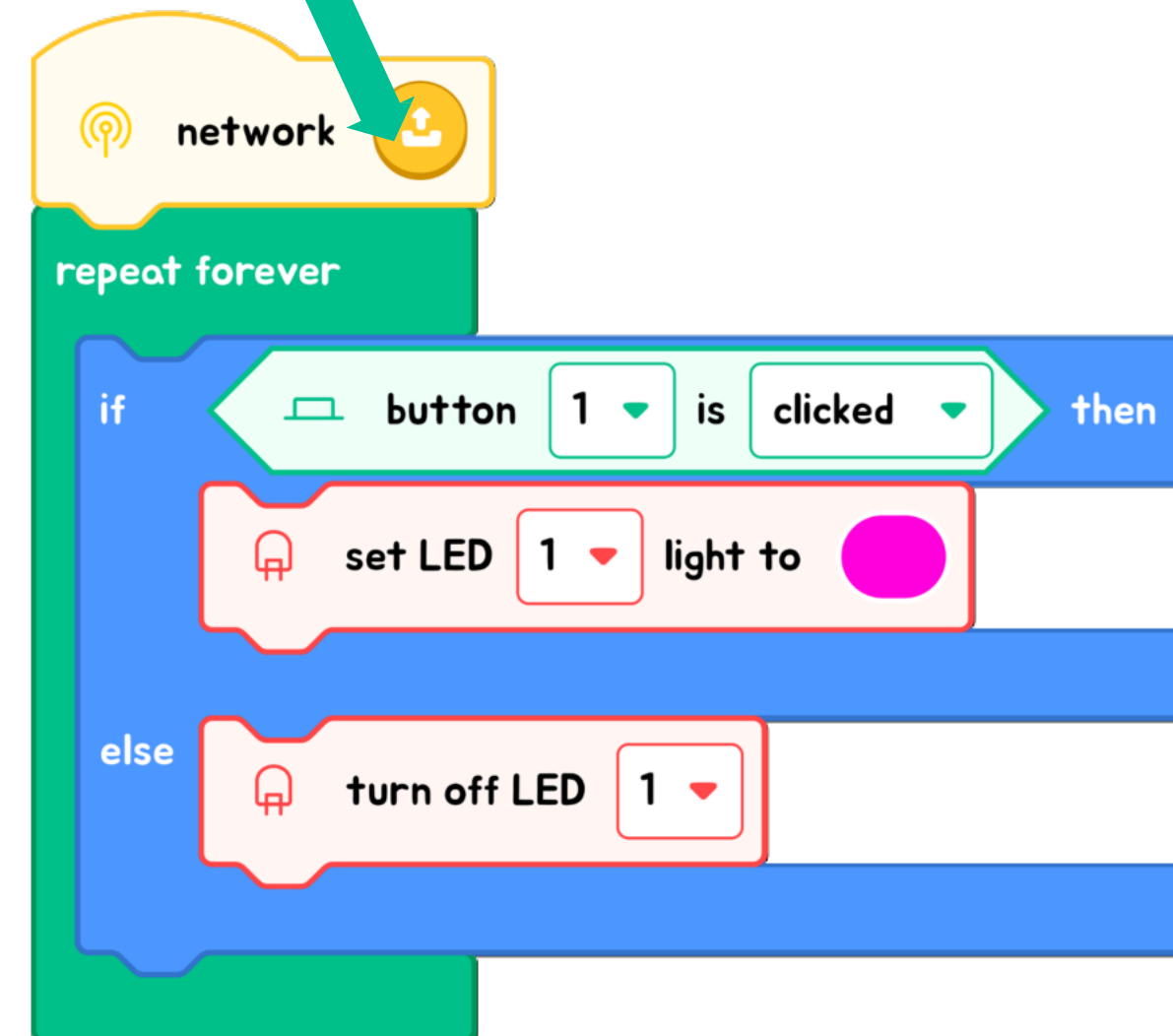
Introduction to MODI – Part.2

PROGRAMMING ACTIVITIES

Step 7 Click on this button!
Then, you can choose your preferred color!



Step 8 Click on this button to upload your code to the modules of the flashlight!



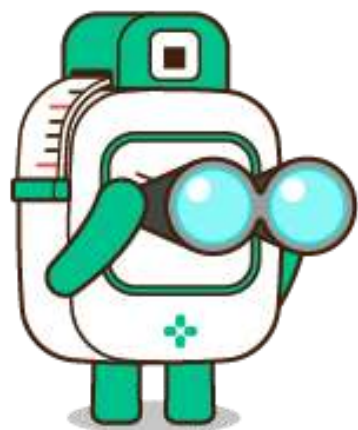
Introduction to Code Editor

Introduction to MODI – Part.2

PROGRAMMING ACTIVITIES

Step 9

After uploading the code successfully, click the button of the flashlight!



What color is the light?

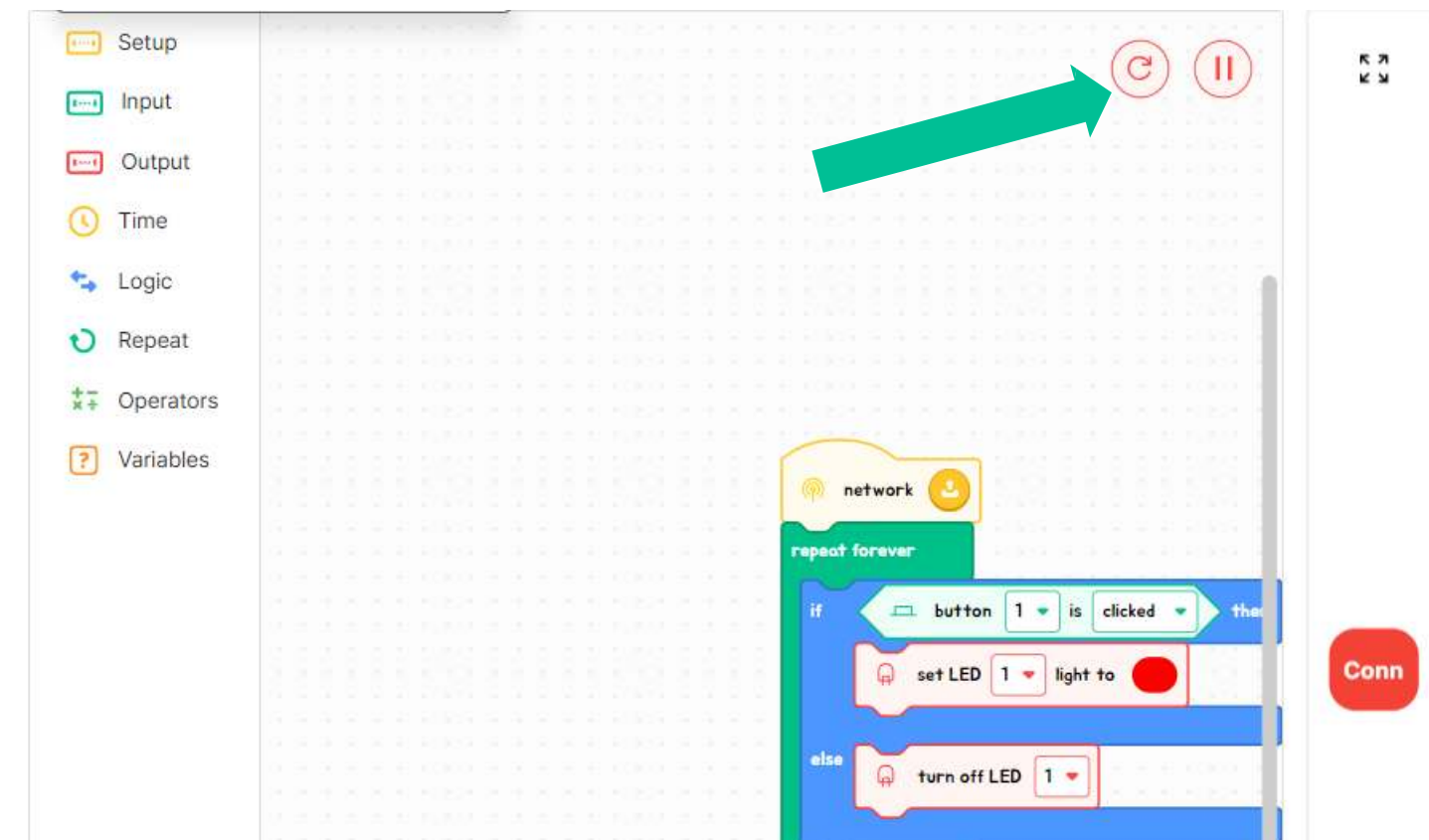
Show your flashlight to your classmates!

Step 10

After completing your project, please reset the modules used for it.



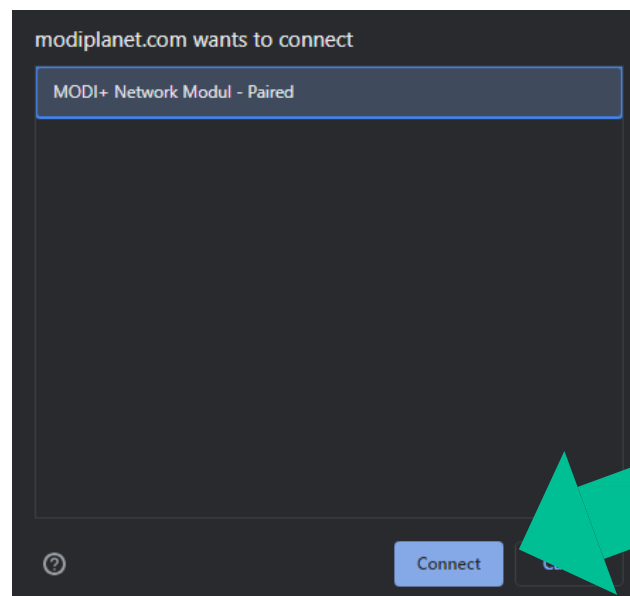
Since the modules retain the uploaded code, they need to be reset for future use !



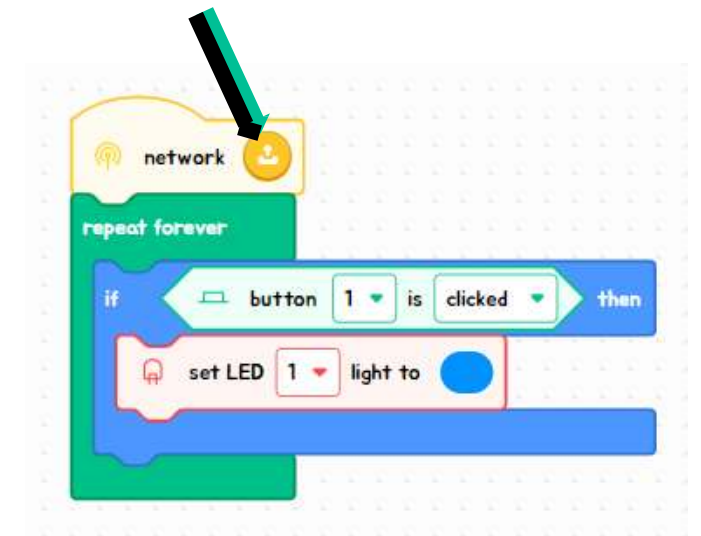
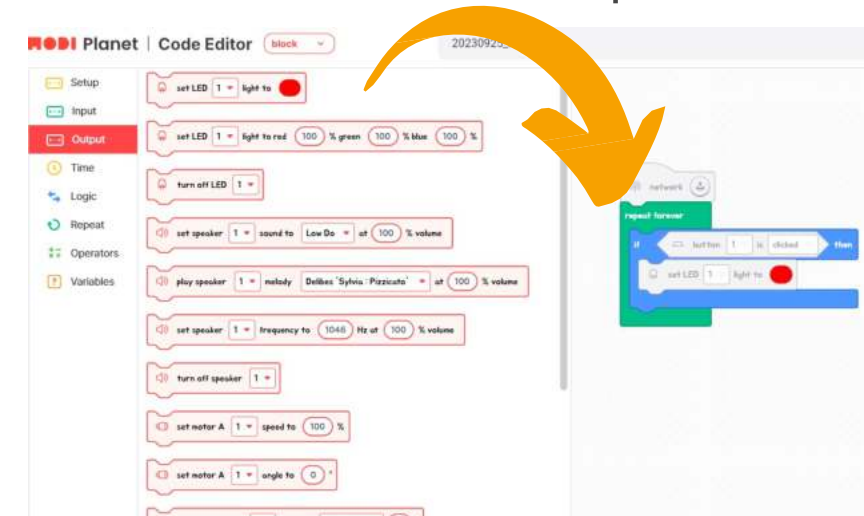
4 Essential Steps to Remember

Step 1 Open Code Editor, and connect the network module to your device (with a type-c cable).

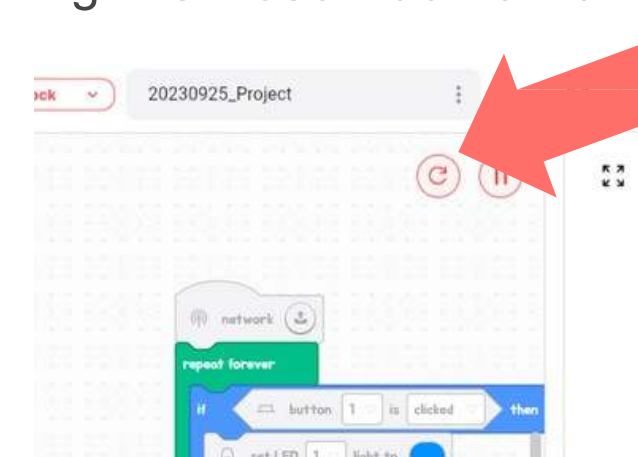
Step 2 Pair the network module in Code Editor



Step 3 Write your code by dragging the required blocks , then upload the code by clicking the upload button on top of the blocks



Step 4 After finishing the project, initialize the modules by clicking the "reset" button on the screen.



Takeaway Points !!

Keep experimenting! Stay curious! and Continue to build on the knowledge you've learned today!!

- ✓ **Input and Output** : *Learned how robots receive (input) and respond (output) to information.*
- ✓ **Basics of Programming** : *Gained foundational knowledge in programming through block coding.*
- ✓ **Electronic Devices** : *Discussed how everyday electronic devices function using basic programming concepts.*
- ✓ **Real-World Applications** : *Linked the concepts to real-world examples of electronic devices around us.*

